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PREHISTORIC ROCK PAINTINGS

THE distant past when there was no paper or language or the written word, and hence no books or written documents, is called prehistory or as we often say, prehistoric times. How people lived in those times was difficult to surmise until scholars began to discover the places where prehistoric people lived. Excavation at these places brought to light old tools, pottery, habitats, bones of ancient human beings and animals, and drawings on cave walls. By piecing together the information deduced from these objects and the cave drawings, scholars have constructed fairly accurate knowledge about what happened and how people lived in prehistoric times. When the basic needs of food, water, clothing and shelter were fulfilled people felt the need to express themselves. Painting and drawing were the oldest art forms practised by human beings to express themselves, using the cave walls as their canvas.



Why did prehistoric people draw these pictures? They may have drawn and painted to make their shelters more colourful and beautiful or to keep a visual record of their day-to-day life, like some of us who maintain a diary.



> The prehistoric period in the early development of human beings is commonly known as the Old Stone Age or the Palaeolithic Age.

Prehistoric paintings have been found in many parts of the world. We do not really know if Lower Palaeolithic people ever produced any art objects. But by the Upper Palaeolithic times we see a proliferation of artistic activities. Around the world the walls of many caves of this time are full of finely carved and painted pictures of animals which the cave-dwellers hunted. The subjects of their drawings were human figures, human activities, geometric designs and animal symbols. In India the earliest paintings have been reported from the Upper Palaeolithic times.



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It is interesting to know that the first discovery of rock paintings was made in India in 1867–68 by an archaeologist, Archibold Carlleyle, twelve years before the discovery of Altamira in Spain. Cockburn, Anderson, Mitra and Ghosh were the early archaeologists who discovered a large number of sites in the Indian sub-continent.

Remnants of rock paintings have been found on the walls of the caves situated in several districts of Madhya Pradesh, Uttar Pradesh, Andhra Pradesh, Karnataka and Bihar. Some paintings have been reported from the Kumaon hills in Uttarakhand also. The rock shelters on banks of the River Suyal at Lakhudiyar, about twenty kilometres on the Almora–Barechina road, bear these prehistoric paintings. Lakhudiyar literally means one lakh caves.The paintings here can be divided into three categories: man, animal and geometric patterns in white, black and red ochre. Humans are represented in stick-like forms. A long-snouted animal, a fox and a multiple legged lizard are the main animal motifs. Wavy lines, rectangle-filled geometric designs, and groups of dots can also be seen here. One of the interesting scenes depicted here is of hand-linked dancing human figures. There is some superimposition of paintings. The earliest are in black; over these are red ochre paintings and the last group comprises white paintings. From Kashmir two slabs with engravings have been reported. The granite rocks of Karnataka and Andhra Pradesh provided suitable canvases to the Neolithic man for his paintings. There are several such sites but more famous among them are Kupgallu, Piklihal and Tekkalkota. Three types of paintings have been reported from here—paintings in white, paintings in red ochre over a white background and paintings in red ochre. These



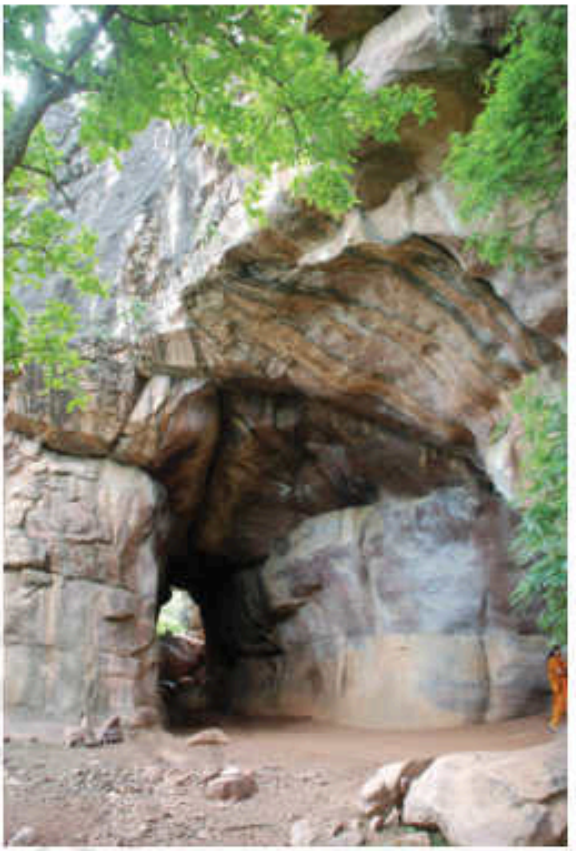
Hand-linked dancing figures, Lakhudiyar, Uttarakhand



Wavy lines, Lakhudiyar, Uttarakhand

paintings belong to late historical, early historical and Neolithic periods. The subjects depicted are bulls, elephants, sambhars, gazelles, sheep, goats, horses, stylised humans, tridents, but rarely, vegetal motifs.

But the richest paintings are reported from the Vindhya ranges of Madhya Pradesh and their Kaimurean extensions into Uttar Pradesh. These hill ranges are full of Palaeolithic and Mesolithic remains, and they are also full of forests, wild plants, fruits, streams and creeks, thus a perfect place for Stone Age people to live. Among these the largest and most spectacular rock-shelter is located in the Vindhya hills at Bhimbetka in Madhya Pradesh. Bhimbetka is located forty-five kilometres south of Bhopal, in an area of ten square kilometres, having about eight hundred rock shelters, five hundred of which bear paintings.



Cave entrance, Bhimbetka, Madhya Pradesh

The caves of Bhimbetka were discovered in 1957–58 by eminent archaeologist V.S. Wakankar and later on many more were discovered. Wakankar spent several years in surveying these inaccessible hills and jungles to study these paintings.

The themes of paintings found here are of great variety, ranging from mundane events of daily life in those times to sacred and royal images. These include hunting, dancing, music, horse and elephant riders, animal fighting, honey collection, decoration of bodies, and other household scenes.

The rock art of Bhimbetka has been classified into various groups on the bases of style, technique and superimposition. The drawings and paintings can be catagorised into seven historical periods. Period I, Upper Palaeolithic; Period II, Mesolithic; and Period III, Chalcolithic. After Period III there are four successive periods. But we will confine ourselves here only to the first three phases.

Can you figure out what the artist is trying to depict in this painting?



Upper Palaeolithic Period

The paintings of the Upper Palaeolithic phase are linear representations, in green and dark red, of huge animal figures, such as bisons, elephants, tigers, rhinos and boars besides stick-like human figures. A few are wash paintings but mostly they are filled with

geometric patterns. The green paintings are of dancers and the red ones of hunters.

Mesolithic Period

The largest number of paintings belong to Period II that covers the Mesolithic paintings. During this period the themes are multiple but the paintings are smaller in size. Hunting scenes predominate. The hunting scenes depict people hunting in groups, armed with barbed spears, pointed sticks, arrows and bows. In some paintings these primitive men are shown with traps and snares probably to catch animals. The hunters are shown wearing simple clothes and ornaments. Sometimes, men have been adorned with elaborate head-dresses, and sometimes painted with masks also. Elephant, bison, tiger, boar, deer, antelope, leopard, panther, rhinoceros, fish, frog, lizard, squirrel and at times birds are also depicted. The Mesolithic artists loved to paint animals. In some pictures, animals are chasing men. In others they are being chased and hunted by men. Some of the animal paintings, especially in the hunting scenes, show a fear of animals, but many others show a feeling of tenderness and love for them. There are also a few engravings representing mainly animals.

Though animals were painted in a naturalistic style, humans were depicted only in a stylistic manner. Women are painted both in the nude and clothed. The young and the old equally find place in these paintings. Children are painted running, jumping and playing. Community dances provide a common theme. There are paintings of people gathering fruit or honey from trees, and of women grinding and preparing food. Some of the pictures of men, women and children seem to depict a sort of family life. In many of the rock-shelters we find hand prints, fist prints, and dots made by the fingertips.

One of the few images showing only one animal, Bhimbetka



The artists of Bhimbetka used many colours, including various shades of white, yellow, orange, red ochre, purple, brown, green and black. But white and red were their favourite colours. The paints were made by grinding various rocks and minerals. They got red from haematite (known as *geru* in India). The green came from a green variety of a stone called chalcedony. White might have been

made out of limestone. The rock of mineral was first ground into a powder. This may then have been mixed with water and also with some thick or sticky substance such as animal fat or gum or resin from trees. Brushes were made of plant fibre. What is amazing is that these colours have survived thousands of years of adverse weather conditions. It is believed that the colours have remained intact because of the chemical reaction of the oxide present on the surface of the rocks.



Painting showing a man being hunted by a beast, Bhimbetka

Why has the animal been shown so big and man so small?

The artists here made their paintings on the walls and ceilings of the rock shelters. Some of the paintings are reported from the shelters where people lived. But some others were made in places which do not seem to have been living spaces at all. Perhaps these places had some religious importance. Some of the most beautiful paintings are very high up on rock shelters or close to the ceilings of rock-shelters. One may wonder why early human beings chose to paint on a rock in such an uncomfortable position. The paintings made at these places were perhaps for people to be able to notice them from a distance.

The paintings, though from the remote past, do not lack pictorial quality. Despite various limitations such as acute working conditions, inadequate tools, materials, etc., there is a charm of simple rendering of scenes of the environment in which the artists lived. The men shown in them appear adventurous and rejoicing in their lives. The animals are shown more youthful and majestic than perhaps they actually were. The primitive artists seem to possess an intrinsic passion for storytelling. These pictures depict, in a dramatic way, both men and animals engaged in the struggle for survival. In one of the scenes, a group of people have been shown hunting a bison. In the process, some injured men are depicted lying scattered on the ground. In another scene, an animal is shown in the agony of death and the men are depicted dancing. These kinds of paintings might have given man a sense of power over the animals he would meet in the open.

This practice is common among primitive people of today also. They engrave or paint on rocks as part of the rituals they perform at birth, at death, at coming of age and at the time of marriage. They dance, masked, during hunting rites to help them kill animals difficult to find or kill.

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HUNTING SCENE



Hunting scenes predominate in Mesolithic paintings. This is one such scene where a group of people are shown hunting a bison. Some injured men are depicted lying scattered on the ground. These paintings show mastery in the skill of drawing these forms.

DANCING SCENE



In this picture hand-linked figures in dancing mode are shown. In fact, this is a recurrent theme. It also recalls the dancing scene from the Lakhudiyar rock painting found in Uttarakhand.

The paintings of individual animals show the mastery of skill of the primitive artist in drawing these forms. Both, proportion and tonal effect, have been realistically maintained in them.

It is interesting to note that at many rock-art sites often a new painting is painted on top of an older painting. At Bhimbetka, in some places, there are as many as 20 layers of paintings, one on top of another. Why did the artists paint in the same place again and again? Maybe, this was because the artist did not like his creation and painted another painting on the previous one, or some of the paintings and places were considered sacred or special or this was because the area may have been used by different generations of people at different times.

These prehistoric paintings help us to understand about early human beings, their lifestyle, their food habits, their daily activities and, above all, they help us understand their mind—the way they thought. Prehistoric period remains are a great witness to the evolution of human civilisation, through the numerous rock weapons, tools, ceramics and bones. More than anything else, the rock paintings are the greatest wealth the primitive human beings of this period left behind.

EXERCISE

1. According to your observation how did the people of prehistoric times select themes for their paintings?
2. What could have been the reasons for depicting more animal figures than human figures in cave paintings?
3. Many visuals of prehistoric cave paintings have been given in this chapter. Among these which one do you like the most and why? Give a critical appreciation of the visual.
4. Other than Bhimbetka, which are the other major sites where these prehistoric paintings have been found? Prepare a report on different aspects of these paintings with pictures or line drawings.
5. In modern times, how have walls been used as a surface to make paintings, graphics, etc?

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and carved standing on a square or circular abacus. Abacuses are decorated with stylised lotuses. Some of the existing pillars with capital figures were found at Basarah-Bakhira, Lauriya-Nandangarh and Rampurva in Bihar, Sankisa and Sarnath in Uttar Pradesh.

The Mauryan pillar capital found at Sarnath popularly known as the Lion Capital is the finest example of Mauryan sculptural tradition. It is also our national emblem. It is carved with considerable care—voluminous roaring lion figures firmly standing on a circular abacus which is carved with the figures of a horse, a bull, a lion and an elephant in vigorous movement, executed with precision, showing considerable mastery in the sculptural techniques. This pillar capital symbolising *Dhammachakrapravartana* (the first sermon by the Buddha) has become a standard symbol of this great historical event in the life of the Buddha.

Monumental images of *Yaksha*, *Yakhinis* and animals, pillar columns with capital figures, rock-cut caves belonging to the third century BCE have been found in different parts of India. It shows the popularity of *Yaksha* worship and how it became part of figure representation in Buddhist and Jaina religious monuments.

Large statues of *Yakshas* and *Yakhinis* are found at many places like Patna, Vidisha and Mathura. These monumental images are mostly in the standing position. One of the distinguishing elements in all these images is their polished surface. The depiction of faces is in full round with pronounced cheeks and physiognomic detail. One of the finest examples is a *Yakshi* figure from Didarganj, Patna, which is tall and well-built. It shows sensitivity towards depicting the human physique. The image has a polished surface.

Yaksha, Parkham



Terracotta figurines show a very different delineation of the body as compared to the sculptures. Depiction of a monumental rock-cut elephant at Dhauli in Odisha shows modelling in round with linear rhythm. It also has Ashokan rock-edict. All these examples are remarkable in their execution of figure representation. The rock-cut cave carved at Barabar hills near Gaya in Bihar is known as the Lomus Rishi cave. The facade of the cave is decorated with the semicircular *chaitya* arch as the entrance. The elephant frieze carved in high relief on the *chaitya* arch shows considerable movement. The interior hall of this cave is rectangular with a circular chamber at the back. The entrance is located on the side wall of the hall. The cave was donated by Ashoka for the Ajivika sect. The Lomus Rishi cave is an example of this period. But many Buddhist caves of the subsequent periods were excavated in eastern and western India.

Due to the popularity of Buddhism and Jainism, *stupas* and *viharas* were constructed on a large scale. However, there are also examples of a few Brahmanical gods in the sculptural representations. It is important to note that the *stupas* were constructed over the relics of the Buddha at Rajagraha, Vaishali, Vethadipa and Pava in Bihar, Kapilavastu, Allakappa and Ramagrama in Nepal, Kushinagar and Pippalvina in Uttar Pradesh. The textual tradition also mentions construction of various other *stupas* on the relics of the Buddha at several places including Avanti and Gandhara which are outside the Gangetic valley.

Stupa, *vihara* and *chaitya* are part of Buddhist and Jaina monastic complexes but the largest number belongs to the Buddhist religion. One of the examples of the structure of a *stupa* in the third century BCE is at Bairat in Rajasthan. The great *stupa* at Sanchi (which will be discussed later) was built with bricks during the time of Ashoka and later it was covered with stone and many new additions were made.

Subsequently many such *stupas* were constructed which shows the popularity of Buddhism. From the second century BCE onwards, we get many inscriptional evidences mentioning donors and, at times, their profession. The pattern of patronage has been a very collective one and there are very few examples of royal patronage. Patrons range from lay devotees to *gahapatis* and kings. Donations by the guilds are also mentioned at several sites. However, there are very few inscriptions mentioning the names of artisans such as Kanha at Pitalkhora and his disciple Balaka at Kondane caves in Maharashtra. Artisans' categories like stone carvers, goldsmiths, stone-polishers, carpenters, etc. are also mentioned in the inscriptions.



Elephant, Dhauli



Lomus Rishi cave-entrance detail

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Stupa *worship, Bharhut*

The method of working was collective in nature and at times only a specific portion of the monument is said to have been patronised by a particular patron. Traders recorded their donation along with their place of origin.

In the subsequent century, *stupas* were elaborately built with certain additions like the enclosing of the circumambulatory path with railings and sculptural decoration. There were numerous *stupas* constructed earlier but expansions or new additions were made in the second century BCE. The *stupa* consists of a cylindrical drum and a circular *anda* with a *harmika* and *chhatra* on the top which remain consistent throughout with minor variations and changes in shape and size. Apart from the circumambulatory path, gateways were added. Thus, with the elaborations in *stupa* architecture, there was ample space for the architects and sculptors to plan elaborations and to carve out images.

During the early phase of Buddhism, Buddha is depicted symbolically through footprints, *stupas*, lotus throne, *chakra*, etc. This indicates either simple worship, or paying respect, or at times depicts historisisation of life events. Gradually narrative became a part of the Buddhist tradition. Thus events from the life of the Buddha, the *Jataka* stories, were depicted on the railings and *torans* of the *stupas*. Mainly synoptic narrative, continuous narrative and episodic narrative are used in the pictorial tradition. While events from the life of the Buddha became an important theme in all the Buddhist monuments, the *Jataka* stories also became equally important for sculptural decorations. The main events associated with the Buddha's life which were frequently depicted were events related to the birth, renunciation, enlightenment, *dhammachakra-pravartana*, and *mahaparinibbana* (liberation from the cycle of birth.) Among the *Jataka* stories that are frequently depicted are Chhadanta *Jataka*, Vidurpundita *Jataka*, Ruru *Jataka*, Sibi *Jataka*, Vessantara *Jataka* and Shama *Jataka*.

EXERCISE

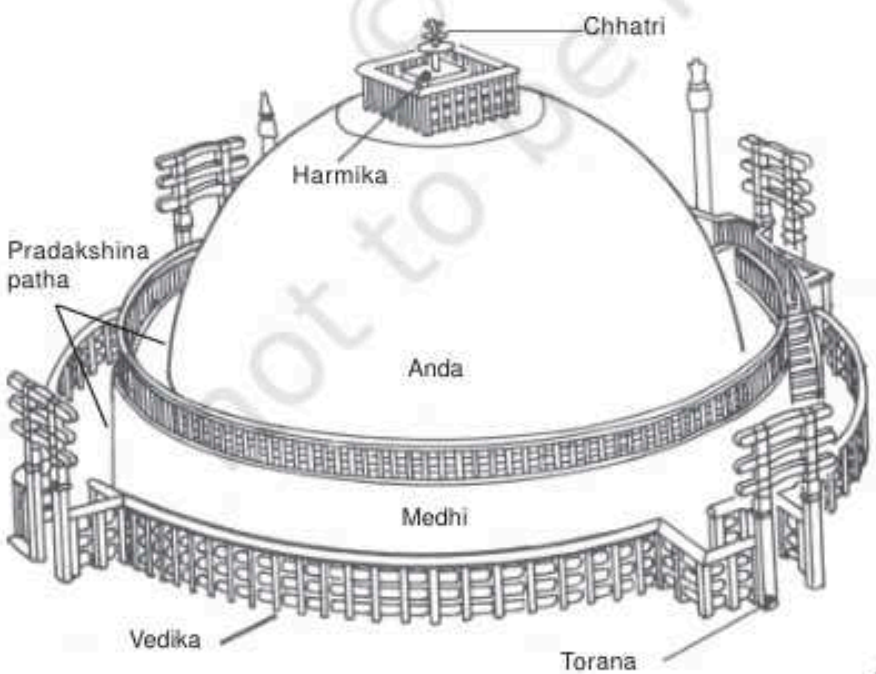
1. Do you think that the art of making sculptures in India began during the Mauryan period?
2. What was the significance of the *stupa* and how did *stupa* architecture develop?
3. Which were the four events in the life of the Buddha which have been depicted in different forms of Buddhist art? What did these events symbolise?
4. What are the *Jatakas*? How do the *Jatakas* relate to Buddhism? Find out.

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rescued by the deer is also shown along with the king pointing a finger at the deer. According to the story, the man promised the deer after his rescue that he would not disclose his identity to anybody. But when the king makes a proclamation of reward for disclosing the identity of the deer, he turns hostile and takes the king to the same jungle where he had seen the deer. Such *Jataka* stories became part of *stupa* decoration. Interestingly, with the rise in the construction of *stupas* in various parts of the country, regional stylistic variations also began to emerge. One main characteristic in all the male images of first–second centuries BCE is the knotted headgear. In many sculptures it is very consistent. Some of the sculptures found at Bharhut are displayed in Indian Museum, Kolkata.

Sanchi

The next phase of sculptural development at Sanchi Stupa-1, Mathura, and Vengi in Andhra Pradesh (Guntur District) is noteworthy in the stylistic progression. Stupa-1 at Sanchi has upper as well as lower *pradakshinapatha* or circumambulatory path. It has four beautifully decorated *toranas* depicting various events from the life of the Buddha and the *Jatakas*. Figure compositions are in high relief, filling up the entire space. Depiction of posture gets naturalistic and there is no stiffness in the body. Heads have considerable projection in the picture space. Rigidity



Plan of Stupa-1, Sanchi

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Stone carving, Stupa-1, Sanchi



Part of railing, Sangol

in the contours gets reduced and images are given movement. Narration gets elaborated. Carving techniques appear more advanced than Bharhut. Symbols continue to be used representing the Buddha. At Sanchi Stupa-1, narratives get more elaborated; however, the depiction of the dream episode remains very simple showing the reclining image of the queen and the elephant at the top. The historical narratives such as the siege of Kushinara, Buddha's visit to Kapilavastu, visit of Ashoka to the Ramgrama Stupa are carved with considerable details. In Mathura, images of this period bear the same quality but are different in the depiction of physiognomic details.

Mathura, Sarnath and Gandhara Schools

The first century CE onwards, Gandhara (now in Pakistan), Mathura in northern India and Vengi in Andhra Pradesh emerged as important centres of art production. Buddha in the symbolic form got a human form in Mathura and Gandhara. The sculptural tradition in Gandhara had the confluence of Baetria, Parthia and the local Gandhara tradition. The local sculptural tradition at Mathura became so strong that the tradition spread to other parts of northern India. The best example in this regard is the *stupa* sculptures found at Sanghol in the Punjab. The Buddha image at Mathura is modelled on the lines of earlier *Yaksha* images whereas in Gandhara it has Hellenistic features. Early Jain Teerthankar images and portraits of kings, specially the headless Kanishka are also found from Mathura.

Images of Vaishnava (mainly Vishnu and his various forms) and Shaiva (mainly the *lingas* and *mukhalingas*) faiths are also found at Mathura but Buddhist images are found in large numbers. It may be noted that the images of Vishnu and Shiva are represented by their *ayudhas* (weapons). There is boldness in carving the large images, the volume of the images is projected out of the picture plane, the faces are round and smiling, heaviness in the sculptural volume is reduced to relaxed flesh. The garments of the body are clearly visible and they cover the left shoulder. Images of the Buddha, *Yakshas*, *Yakshinis*, Shaivite and Vaishnavite deities and portrait statues are profusely sculpted. In the second century CE, images in Mathura get sensual, rotundity increases, they become fleshier. The trend continues in the fourth century CE but in the late fourth century CE, the massiveness and fleshiness is reduced further and the flesh becomes more tightened, the volume of the drapery also gets reduced and in the fifth and sixth centuries CE, the drapery is

integrated into the sculptural mass. Transparent quality in the robes of the Buddha images is evident. In this period, two important schools of sculptures in northern India are worth noting. The traditional centre, Mathura, remained the main art production site whereas Sarnath and Kosambi also emerged as important centres of art production. Many Buddha images in Sarnath have plain transparent drapery covering both shoulders, and the halo around the head has very little ornamentation whereas the Mathura Buddha images continue to depict folds of the drapery in the Buddha images and the halo around the head is profusely decorated. One can visit museums at Mathura, Sarnath, Varanasi, New Delhi, Chennai, Amaravati, etc. to study the features of early sculptures.

Among the important *stupa* sites outside the Gangetic valley is Devnimori in Gujarat. In the subsequent centuries sculptures had little variations while slender images with transparent drapery remained a dominant aesthetic sensibility.



Meditating Buddha, Gandhar, third–fourth century CE



Bodhisattva, Gandhar, fifth–sixth century CE

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Buddhist Monuments of South India

Vengi in Andhra Pradesh has many *stupa* sites like Jagayyapetta, Amaravati, Bhattiprolu, Nagarjunkonda, Goli, etc. Amaravati has a *mahachaitya* and had many sculptures which are now preserved in Chennai Museum, Amaravati Site Museum, National Museum, New Delhi and the British Museum, London. Like the Sanchi Stupa, the Amaravati Stupa also has *pradakshinapatha* enclosed within a *vedika* on which many narrative sculptures are depicted. The domical *stupa* structure is covered with relief *stupa* sculptural slabs which is a unique feature. The *torana* of the Amaravati Stupa has disappeared over a period of time. Events from the life of the Buddha and the *Jataka* stories are depicted. Though in the Amaravati Stupa there is evidence of construction activity in the third century BCE, it was best developed in the first and second centuries CE. Like Sanchi, the early phase is devoid of Buddha images but during the later phase, in the second and third centuries CE, the Buddha images are carved on the drum slabs and at many other places. Interior space in the composition is created by different postures of the figures such as semi-back, back, profile, frontal, semi-frontal, side, etc.

Sculptural form in this area is characterised by intense emotions. Figures are slender, have a lot of movement, bodies are shown with three bents (i.e. *tribhanga*), and the sculptural composition is more complex than at Sanchi. Linearity becomes flexible, dynamic movement breaks the



Carving on outer wall of Stupa, Amaravati



Stupa drum slab, Amravati, second century CE

staticness of form. The idea of creating three-dimensional space in the relief sculpture is devised by using pronounced volume, angular bodies and complex overlapping. However, absolute attention has been paid to the clarity of form despite its size and role in the narrative. Narratives are profusely depicted which include events from the life of the Buddha and the *Jataka* stories. There are a number of *Jataka* scenes that have not been completely identified. In the depiction of the birth event, the queen is shown reclining on a bed surrounded by female attendants and a small-sized elephant is carved on the upper frame of the composition showing the dream of Queen Mayadevi. In another relief, four events related to the birth of the Buddha are shown. These represent varied ways of depicting the narratives.

The animated movement in the figures gets reduced in the sculptures of Nagarjunkonda and Goli in the third century CE. Even within the relatively low relief volume than in the Amaravati sculptures, artists at Nagarjunkonda and Goli managed to create the effect of protruding surfaces of the body which is suggestive in nature and look very integral. Independent Buddha images are also found at Amaravati, Nagarjunkonda and Guntapalle in Andhra Pradesh. Guntapalle is a rock-cut cave site near Eluru. Small apsidal and circular *chaitya* halls have been excavated belonging to the second century BCE. The other important site where rock-cut *stupas* have been excavated is Anakapalle near Vishakhapatanam. In Karnataka, Sannati in Gulbarga district is the largest *stupa* site excavated so far. It also has a *stupa* like the one in Amaravati decorated with sculptural relief.



Panel, Nagarjunkonda

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Construction of a large number of stupas does not mean that there were no structured temples or *viharas* or *chaityas*. We do get evidences but no structured *chaitya* or *vihara* survived. Among the important structured *viharas*, mention may be made of the Sanchi apsidal *chaitya* structure, i.e., temple 18, which is a simple shrine temple having front pillars and a hall at the back. Similar structured temples at Guntapalle are also worth mentioning. Along with the images of the Buddha, other Buddhist images of Boddhisattvas like Avalokiteshvara, Padmapani, Vajrapani, Amitabha, and Maitreya Buddha started getting sculpted. However, with the rise of *Vajrayana* Buddhism many Boddhisattva images were added as a part of the personified representations of certain virtues or qualities as propagated by the Buddhist religious principles for the welfare of the masses.

Cave Tradition in Western India

In western India, many Buddhist caves dating back to the second century BCE onwards have been excavated. Mainly three architectural types were executed—(i) apsidal vault-roof *chaitya* halls (found at Ajanta, Pitalkhora, Bhaja); (ii) apsidal vault-roof pillarless hall (found at Thana-Nadsur in Maharashtra); and (iii) flat-roofed quadrangular hall with a circular chamber at the back (found at Kondivite in Maharashtra). The front of the *chaitya* hall is dominated by the motif of a semi-circular *chaitya* arch with an open front which has a wooden facade and, in some cases, there is no dominating *chaitya* arch window such as found at Kondivite. In all the *chaitya* caves a stupa at the back is common.

In the first century BCE some modifications were made to the standard plan of the apsidal vault-roof variety where the hall becomes rectangular like at Ajanta Cave No. 9



Unfinished chaitya cave, Kanheri



Chaitya hall, Karla

with a stone-screen wall as a facade. It is also found at Bedsa, Nashik, Karla and Kanheri. Many cave sites have the standard first type of *chaitya* halls in the subsequent period. In Karla, the biggest rock-cut *chaitya* hall was excavated. The cave consists of an open courtyard with two pillars, a stone screen wall to protect from rain, a veranda, a stone-screen wall as facade, an apsidal vault-roof *chaitya* hall with pillars, and a *stupa* at the back. Karla *chaitya* hall is decorated with human and animal figures. They are heavy in their execution, and move in the picture space. Further elaboration over the Karla *chaitya* hall plan is observed at Kanheri Cave No.3. Though the cave's interior was not fully finished, it shows how the carving progressed from time to time. Subsequently, the quadrangular flat-roofed variety became the most preferred design and is extensively found at many places.



Nashik Cave No. 3

The *viharas* are excavated in all the cave sites. The plan of the *viharas* consists of a veranda, a hall and cells around the walls of the hall. Some of the important *vihara* caves are Ajanta Cave No. 12, Bedsa Cave No. 11, Nashik Cave Nos. 3, 10 and 17. Many of the early *vihara* caves are carved with interior decorative motifs like *chaitya* arches and the *vedica* designs over the cell doors of the cave. Facade design in Nashik Cave Nos. 3, 10, and 17 became a distinct achievement. The *vihara* caves at Nashik were excavated with front pillars carved with *ghata*-base and *ghata*-capital with human figures. One such *vihara* cave was also excavated at Junnar in Maharashtra, which is popularly known as Ganeshleni because an image of Ganesha belonging to a later period was installed in it. Later, a stupa was added at the back of the hall of the *vihara* and it became a *chaitya-vihara*. The stupas in the fourth and fifth centuries CE have Buddha images attached. Junnar has the largest cave excavations— more than two hundred caves around the hills of the town— whereas Kanheri in Mumbai has a hundred and eight excavated caves. The

Chaitya, Cave No. 12, Bhaja



most important sites are Ajanta, Pitalkhora, Ellora, Nashik, Bhaja, Junnar, Karla, Kanheri. Ajanta, Ellora, and Kanheri continue to flourish.

Ajanta

The most famous cave site is Ajanta. It is located in Aurangabad District of Maharashtra State. Ajanta has twenty-nine caves. It has four *chaitya* caves datable to the earlier phase, i.e., the second and the first century BCE (Cave Nos. 10 and 9) and the later phase, i.e., the fifth century CE (Cave Nos. 19 and 26). It has large *chaitya-viharas* and is decorated with sculptures and paintings. Ajanta is the only surviving example of painting of the first century BCE and the fifth century CE. The caves at Ajanta as well as in western Deccan in general have no precise chronology because of the lack of known dated inscriptions.

Cave Nos. 10, 9, 12 and 13 belong to the early phase, Cave Nos. 11, 15 and 6 upper and lower, and Cave No. 7 belong to the phase earlier than late fifth century CE. The rest of the caves belong to late fifth century CE to early sixth century CE. The *chaitya* Cave Nos. 19 and 26 are elaborately carved. Their facade is decorated with Buddha and, Boddhisattva images. They are of the apsidal-vault-roof variety. Cave No. 26 is very big and the entire interior hall is carved with a variety of Buddha images, the biggest one being the *Mahaparinibbana* image. The rest of the caves are *vihara-chaitya* caves. They consist of a pillared veranda, a pillared hall and cells along the walls. The back wall has the main Buddha shrine. Shrine images at Ajanta are grand in size. Some of the *vihara* caves are unfinished such as Cave Nos. 5, 14, 23 24, 28 and 29. Among the important patrons at Ajanta were Varahadeva (patron of Cave No. 16), the prime minister of the Vakataka king, Harishena; Upendragupta (patron of Cave Nos. 17–20) the local king of the region and feudatory of the Vakataka king, Harishena; Buddhabhadra (patron of Cave No. 26); and Mathuradasa (patron of Cave No. 4). Many paintings have survived in Cave Nos. 1, 2, 16 and 17.

View, Ajanta Caves



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Part of Mahajanaka Jataka panel, Cave No. 1, Ajanta

Chaddanta faithfully follows the Pali text whereas the one painted in Cave No. 17 is very different. In one of the events, the Boddhisattva, *Chaddanta*, is shown removing his own tusk and giving it to the hunter, Sonuttar. The other important paintings are the famous Padmapani and Vajrapani in Cave No. 1. However, it may be observed that the images of Padmapani and Vajrapani are very common in Ajanta but the best preserved paintings are in Cave No. 1. Some figures in Cave No. 2 have affiliation with the Vengi sculptures and at the same time, the influence of the Vidarbha sculptural tradition is also observed in the delineation of some sculptures. The subsequent development of the painting tradition has been discussed in the next chapter.

Ellora

Another important cave site located in Aurangabad District is Ellora. It is located a hundred kilometres from Ajanta and has thirty-four Buddhist, Brahmanical and Jain caves. It is a unique art-historical site in the country as it has monastries associated with the three religions dating from the fifth century CE onwards to the eleventh century CE. It is also unique in terms of stylistic eclecticism, i.e., confluence of many styles at one place. The caves of Ellora and Aurangabad show the ongoing differences between the two religions—Buddhism and Brahmanical. There are twelve Buddhist caves having many images belonging to Vajrayana Buddhism like Tara, Mahamayuri, Akshobhya, Avalokiteshwara, Maitrya, Amitabha, etc. Buddhist caves



Courtyard, Kailash Temple, Cave No.16, Ellora



Seated Buddha, Chaitya Hall, Cave No. 10, Ellora

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are big in size and are of single, double and triple storeys. Their pillars are massive. Ajanta also has excavated double-storeyed caves but at Ellora, the triple storey is a unique achievement. All the caves were plastered and painted but nothing visible is left. The shrine Buddha images are big in size; they are generally guarded by the images of Padmapani and Vajrapani. Cave No. 12, which is a triple-storey excavation, has images of Tara, Avalokiteshwara, Manushi Buddhas and the images of Vairochana, Akshobhya, Ratnasambhava, Amitabha, Amoghsiddhi, Vajrasatva and Vajraraja. On the other hand, the only double-storey cave of the Brahmanical faith is Cave No. 14. Pillar designs grow from the Buddhist caves and when they reach the Jain caves belonging to the ninth century CE, they become very ornate and the decorative forms gain heavy protrusion.

The Brahmanical Cave Nos. 13–28 have many sculptures. Many caves are dedicated to Shaivism, but the images of both Shiva and Vishnu and their various forms according to Puranic narrative are depicted. Among the Shaivite themes, Ravana shaking Mount Kailash, Andhakasurvadha, Kalyanasundara are profusely depicted whereas among the Vaishnavite themes, the different *avatars* of Vishnu are depicted. The sculptures at Ellora



Gajasur Shiva, Cave No.15, Ellora

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are monumental, and have protruding volume that create deep recession in the picture space. The images are heavy and show considerable sophistication in the handling of sculptural volume. Various guilds at Ellora came from different places like Vidarbha, Karnataka and Tamil Nadu and carved the sculptures. Thereby it is the most diverse site in India in terms of the sculptural styles. Cave No. 16 is known as Kailash leni. A rock-cut temple has been carved out of a single rock, a unique achievement of the artisans, which will be discussed in the next chapter. Among the important Shaivite caves are Cave No. 29 and Cave No. 21. The plan of Cave No. 29 is almost like that of the main cave at Elephanta. The sculptural quality of Cave Nos. 29, 21, 17, 14 and 16 is amazing for its monumentality and vigorous movements in the picture space.

The Bagh Caves, consisting of Buddhist mural paintings, are located 97 km from Dhar district of Madhya Pradesh. These rock-cut cave monuments are not natural but carved over a period of time during the ancient India mostly during the Satvahana period. The Bagh caves, like those at Ajanta, were excavated by master craftsmen on perpendicular sandstone rock face of a hill across the seasonal stream of Baghani. Out of the original nine caves, only five have survived, all of which are *viharas* or resting places for monks, having a quadrangular plan. A small chamber, usually at the back, forms the *Chatiya* — the prayer hall. Most significant of the five caves is Cave No. 4, commonly known as the *Rang Mahal*, meaning the Palace of Colours, where paintings on the wall and ceilings are still visible. Other caves where remanants of tempra mural paintings on the walls and ceiling can be seen are Cave Nos. 2, 3, 5 and 7. The ground prepared was a reddish-brown gritty and thick mud plaster, laid out on the walls and ceilings. Over the plaster, lime-priming was done, on which these paintings were made. Some of the most beautiful paintings were on the walls of the portico of Cave 4. To prevent further loss of the values of the Indian art, most paintings were carefully removed in 1982 and today can be seen in the Archaeological Museum of Gwalior.

Elephanta Caves and Other Sites

The Elephanta Caves located near Mumbai are dominated by the Shaivite faith. It is contemporary with Ellora, and its sculptures show slenderness in the body, with stark light and dark effects.

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Entrance to Elephanta Caves

The tradition of rock-cut caves continued in the Deccan and they are found not only in Maharashtra but also in Karnataka, mainly at Badami and Aiholi, executed under the patronage of the Chalukyas; in Andhra Pradesh in the area of Vijayawada; and in Tamil Nadu, mainly at Mahabalipuram, under the patronage of the Pallavas. The post-sixth-century development of art history in the country depended more on political patronage than the collective public patronage of the early historic periods.

Mention may also be made of the terracotta figurines that are found at many places all over the country. They show a parallel tradition with the religious lithic sculptures as well as the independent local tradition. Many terracotta figures of various sizes are found which show their popularity. They are toys, religious figurines as well as figurines made for healing purposes as part of the belief systems.

Cave Tradition in Eastern India

Like in western India, Buddhist caves have also been excavated in eastern India, mainly in the coastal region of Andhra Pradesh, and in Odisha. One of the main sites in Andhra Pradesh is Guntapalle in Eluru district. The caves have been excavated in the hills along with the structured monasteries. Perhaps it is among the very unique sites where the structured *stupas*, *viharas* and the caves are

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Udaigiri-Khandagiri caves near Bhubaneswar



Details of the veranda, Udaigiri-Khandagiri

excavated at one place. The Guntapalle *chaitya* cave is circular with a *stupa* in the circular hall and a *chaitya* arch carved at the entrance. The cave is relatively small when compared to the caves in western India. A number of *vihara* caves have been excavated. The main *vihara* caves, despite the small dimensions, have been decorated with *chaitya* arches on the exterior. They are rectangular with vaulted roof and are carved single-storeyed or double-storeyed without a large central hall. These excavations date back to the second century BCE. There are some excavations which were added in the subsequent centuries but all are of the *vihara* type. Apart from Guntapalle, the other important cave site is Rampaerrampallam which has very moderate small excavations but there are rock-cut stupas on the hillock. At Anakapalli near Vishakhapatnam, caves were excavated and a huge rock-cut stupa was carved out of the hillock during the fourth–fifth centuries CE. It is a unique site as it has the biggest rock-cut stupas in the country. Many votive rock-cut stupas all around the hillock have also been excavated.

The rock-cut cave tradition also existed in Odisha. The earliest examples are the Udaigiri-Khandagiri caves in the vicinity of Bhubaneswar. These caves are scattered and have inscriptions of Kharavela Jain kings. According to the inscriptions, the caves were meant for Jain monks. There are numerous single-cell excavations. Some have been carved in huge independent boulders and given the shape of animals. The big caves include a cave with a pillared veranda with cells at the back. The upper part of the cells is decorated with a series of *chaitya* arches and

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6

TEMPLE ARCHITECTURE AND SCULPTURE

MOST of the art and architectural remains that survive from Ancient and Medieval India are religious in nature. That does not mean that people did not have art in their homes at those times, but domestic dwellings and the things in them were mostly made from materials like wood and clay which have perished. This chapter introduces us to many types of temples from India. Although we have focussed mostly on Hindu temples, at the end of the chapter you will find some information on major Buddhist and Jain temples too. However, at all times, we must keep in mind that religious shrines were also made for many local cults in villages and forest areas, but again, not being of stone the ancient or medieval shrines in those areas have also vanished.

Today when we say 'temple' in English we generally mean a *devalaya, devkula mandir, kovil, deol, devasthanam or prasada* depending on which part of India we are in.

Early Temples

While construction of *stupas* continued, Brahmanical temples and images of gods also started getting constructed. Often temples were decorated with the images of gods. Myths mentioned in the *Puranas* became part of narrative representation of the Brahmanical religion. Each temple had a principal image of a god. The shrines of the temples were of three kinds—(i) *sandhara* type (with *pradikshinapatha*), (ii) *nirandhara* type (without *pradakshinapatha*), and (iii) *sarvatobhadra* (which can be accessed from all sides). Some of the important temple sites of this period are Deogarh in Uttar



Chatur Mukhlinga, Nachna- Kuthara (Inset)

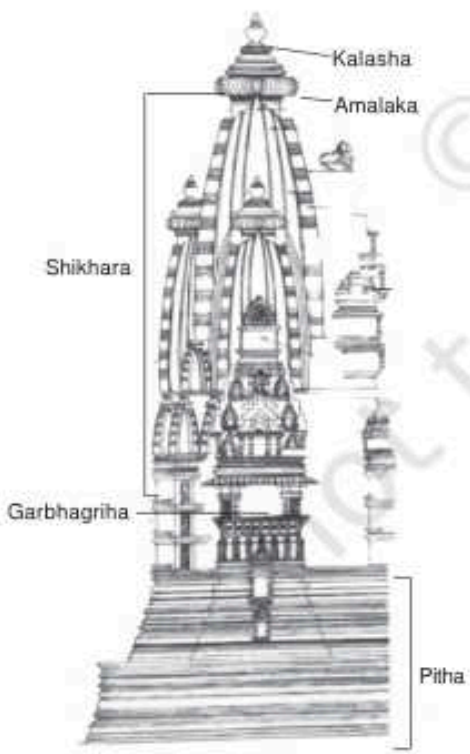
Shiva temple, Nachna-Kuthara, Madhya Pradesh, fifth century CE

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Pradesh, Eran, Nachna-Kuthara and Udaygiri near Vidisha in Madhya Pradesh. These temples are simple structures consisting of a veranda, a hall and a shrine at the rear.

THE BASIC FORM OF THE HINDU TEMPLE

The basic form of the Hindu temple comprises the following: (i) sanctum (*garbhagriha* literally 'womb-house'), which was a small cubicle with a single entrance and grew into a larger chamber in time. The *garbhagriha* is made to house the main icon which is itself the focus of much ritual attention; (ii) the entrance to the temple which may be a portico or colonnaded hall that incorporates space for a large number of worshippers and is known as a *mandapa*; (iii) freestanding temples tend to have a mountain-like spire, which can take the shape of a curving *shikhar* in North India and a pyramidal tower, called a *vimana*, in South India; (iv) the *vahan*, i.e., the mount or vehicle of the temple's main deity along with a standard pillar or *dhvaj* is placed axially before the sanctum. Two broad orders of temples in the country are known— *Nagara* in the north and *Dravida* in the south. At times, the *Vesar* style of temples as an independent style created through the selective mixing of the *Nagara* and *Dravida* orders is mentioned by some scholars. Elaborate studies are available on the various sub-styles within these orders. We will look into the differences in the forms further on in this chapter. As temples grew more complex, more surfaces were created for sculpture through additive geometry, i.e., by adding more and more rhythmically projecting, symmetrical walls and niches, without breaking away from the fundamental plan of the shrine.



Nagara temple

SCULPTURE, ICONOGRAPHY AND ORNAMENTATION

The study of images of deities falls within a branch of art history called 'iconography', which consists of identification of images based on certain symbols and mythologies associated with them. And very often, while the fundamental myth and meaning of the deity may remain the same for centuries, its specific usage at a spot can be a response to its local or immediate social, political or geographical context.

Every region and period produced its own distinct style of images with its regional variations in iconography. The temple is covered with elaborate sculpture and ornament that form a fundamental part of its conception. The placement of an image in a temple is carefully planned:

for instance, river goddesses (Ganga and Yamuna) are usually found at the entrance of a *garbhagriha* in a *Nagara* temple, *dvarapalas* (doorkeepers) are usually found on the gateways or *gopurams* of *Dravida* temples, similarly, *mithunas* (erotic images), *navagrahas* (the nine auspicious planets) and *yakshas* are also placed at entrances to guard them. Various forms or aspects of the main divinity are to be found on the outer walls of the sanctum. The deities of directions, i.e., the *ashtadikpalas* face the eight key directions on the outer walls of the sanctum and/or on the outer walls of a temple. Subsidiary shrines around the main temple are dedicated to the family or incarnations of the main deity. Finally, various elements of ornamentation such as *gavaksha*, *vyala/yali*, *kalpa-lata*, *amalaka*, *kalasha*, etc. are used in distinct ways and places in a temple.

THE *NAGARA* OR NORTH INDIAN TEMPLE STYLE

The style of temple architecture that became popular in northern India is known as *nagara*. In North India it is common for an entire temple to be built on a stone platform with steps leading up to it. Further, unlike in South India it does not usually have elaborate boundary walls or gateways. While the earliest temples had just one tower, or *shikhara*, later temples had several. The *garbhagriha* is always located directly under the tallest tower.

There are many subdivisions of *nagara* temples depending on the shape of the *shikhara*. There are different names for the various parts of the temple in different parts



Sun temple, Konark

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of India; however, the most common name for the simple *shikhara* which is square at the base and whose walls curve or slope inward to a point on top is called the 'latina' or the *rekha-prasada* type of shikara.

The second major type of architectural form in the *nagara* order is the *phamsana*. *Phamsana* buildings tend to be broader and shorter than latina ones. Their roofs are composed of several slabs that gently rise to a single point over the centre of the building, unlike the latina ones which look like sharply rising tall towers. *Phamsana* roofs do not curve inward, instead they slope upwards on a straight incline. In many North Indian temples you will notice that the *phamsana* design is used for the *mandapas* while the main *garbhagriha* is housed in a latina building. Later on, the latina buildings grew complex, and instead of appearing like a single tall tower, the temple began to support many smaller towers, which were clustered together like rising mountain-peaks with the tallest one being in the centre, and this was the one which was always above the *garbhagriha*.

The third main sub-type of the *nagara* building is what is generally called the *valabhi* type. These are rectangular buildings with a roof that rises into a vaulted chamber. The edge of this vaulted chamber is rounded, like the bamboo or wooden wagons that would have been drawn by bullocks in ancient times. They are usually called 'wagon-vaulted buildings'. As mentioned above, the form of the temple is influenced by ancient building forms that were already in existence before the fifth century CE. The *valabhi* type of building was one of them. For instance, if you study



Dashavtara Vishnu temple, Deogarh, fifth century CE



Sheshashayana Vishnu, Dashavatara temple, Deogarh

the ground-plan of many of the Buddhist rock-cut *chaitya* caves, you will notice that they are shaped as long halls which end in a curved back. From the inside, the roof of this portion also looks like a wagon-vaulted roof.

Central India

Ancient temples of Uttar Pradesh, Madhya Pradesh and Rajasthan share many traits. The most visible is that they are made of sandstone. Some of the oldest surviving structural temples from the Gupta Period are in Madhya Pradesh. These are relatively modest-looking shrines each having four pillars that support a small *mandapa* which looks like a simple square porch-like extension before an equally small room that served as the *garbhagriha*. Importantly, of the two such temples that survive, one is at Udaigiri, which is on the outskirts of Vidisha and is part of a larger Hindu complex of cave shrines, while the other one is at Sanchi, near the stupa. This is the first temple having a flat roof. This means that similar developments were being incorporated in the architecture of temples of both the religions.

Deogarh (in Lalitpur District, Uttar Pradesh) was built in the early sixth century CE. That is, about a hundred years or so after the small temples we just learnt about in Sanchi and Udaigiri. This makes it a classic example of a late Gupta Period type of temple. This temple is in the *panchayatana* style of architecture where the main shrine is built on a rectangular plinth with four smaller subsidiary shrines at the four corners (making it a total number of five shrines, hence the name, *panchayatana*). The tall and curvilinear *shikhara* also corroborates this date. The presence of this curving latina or *rekha-prasada* type of *shikhara* also makes it clear that this is an early example of a classic *nagara* style of temple.

> ***Sheshashayana*** is the form of Vishnu where he is shown reclining on the *sheshanaga* called Ananta. ***Nara-Narayan*** shows the discussion between the human soul and the eternal divine. ***Gajendramoksha*** is the story of achieving *moksha*, symbolically communicated by Vishnu's suppression of an *asura* who had taken the form of an elephant.

This west-facing temple has a grand doorway with standing sculptures of female figures representing the Ganga on the left side and the Yamuna on the right side. The temple depicts Vishnu in various forms, due to which it was assumed that the four subsidiary shrines must also

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Vishvanatha temple, Khajuraho

have housed Vishnu's *avatars* and the temple was mistaken for a *dasavatara* temple. In fact, it is not actually known to whom the four subsidiary shrines were originally dedicated. There are three main reliefs of Vishnu on the temple walls: *Sheshashayana* on the south, *Nara-Narayan* on the east and *Gajendramoksha* on the west. The temple is west-facing, which is less common, as most temples are east- or north-facing.

Numerous temples of smaller dimensions have been constructed over a period of time. By contrast, if we study the temples of Khajuraho made by the Chandela Kings in the tenth century, i.e., about four hundred years after the temple at Deogarh, we can see how dramatically the shape and style of the *nagara* temple architecture had developed.

The Lakshmana temple of Khajuraho, dedicated to Vishnu, was built in 954 by the Chandela king, Dhanga. A *nagara* temple, it is placed on a high platform accessed by stairs. There are four smaller temples in the corners, and all the towers or *shikharas* rise high, upward in a curved pyramidal fashion, emphasising the temple's vertical thrust ending in a horizontal fluted disc called an *amalak* topped with a *kalash* or vase. The crowning elements: *amalak* and *kalash*, are to be found on all *nagara* temples of this period. The temple also has projecting balconies and verandahs, thus very different from Deogarh.

Kandariya Mahadeo temple, Khajuraho



Kandariya Mahadeo temple at Khajuraho is the epitome of temple architecture in Central India. In the architecture and the sculptures of this temple, which is a massive structure, we see all features of central Indian temples of the medival period for which they are known and appreciated all over. Khajuraho's temples are also known for their extensive erotic sculptures; the erotic expression is given equal importance in human experience as spiritual pursuit, and it is seen as part of a larger cosmic whole. Many Hindu temples, therefore, feature *mithun* (embracing couple) sculptures, considered auspicious. Usually, they are placed at the entrance of the temple or on an exterior



Dance class, Lakshmana Temple, Khajuraho

wall or they may also be placed on the walls between the *mandapa* and the main shrine. Khajuraho's sculptures are highly stylised with typical features: they are in almost full relief, cut away from the surrounding stone, with sharp noses, prominent chins, long slanting eyes and eyebrows.

There are many temples at Khajuraho, most of them devoted to Hindu gods. There are some Jain temples as well as a Chausanth Yogini temple, which is of interest. Predating the tenth century, this is a temple of small, square shrines of roughly-hewn granite blocks, each dedicated to *devis* or goddesses associated with the rise of *Tantric* worship after the seventh century. Several such temples were dedicated to the cult of the *yoginis* across Madhya Pradesh, Odisha and even as far south as Tamil Nadu. They were built between the seventh and tenth centuries, but few have survived.

West India

The temples in the north-western parts of India including Gujarat and Rajasthan, and stylistically extendable, at times, to western Madhya Pradesh are too numerous to include



Sun temple, Modhera, Gujarat

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Sun temple, Modhera, Gujarat

here in any comprehensive way. The stone used to build the temples ranges in colour and type. While sandstone is the commonest, a grey to black basalt can be seen in some of the tenth to twelveth century temple sculptures. The most exuberant and famed is the manipulatable soft white marble which is also seen in some of the tenth to twelveth century Jain temples in Mount Abu and the fifteenth century temple at Ranakpur.

Among the most important art-historical sites in the region is Samlaji in Gujarat which shows how earlier artistic traditions of the region mixed with a post-Gupta style and gave rise to a distinct style of sculpture. A large number of sculptures made of grey schist have been found in this region which can be dated between the sixth and eighth centuries CE. While the patronage of these is debated, the date is established on the basis of the style.

The Sun temple at Modhera dates back to early eleventh century and was built by Raja Bhimdev I of the Solanki Dynasty in 1026. There is a massive rectangular stepped tank called the *surya kund* in front of it. Proximity of sacred architecture to a water body such as a tank, a river or a pond has been noticed right from the earliest times. By the early eleventh century they had become a part of many temples. This hundred-square-metre rectangular pond is perhaps the grandest temple tank in India. A hundred and eight miniature shrines are carved in between the steps inside the tank. A huge ornamental arch-*torana* leads one to the *sabha mandapa* (the assembly hall) which is open on all sides, as was the fashion of the times in western and central Indian temples.

The influence of the woodcarving tradition of Gujarat is evident in the lavish carving and sculpture work. However, the walls of the central small shrine are devoid of carving and are left plain as the temple faces the east and, every year, at the time of the equinoxes, the sun shines directly into this central shrine.

East India

Eastern Indian temples include those found in the North-East, Bengal and Odisha. Each of these three areas produced distinct types of temples. The history of architecture in the North-East and Bengal is hard to study because a number of ancient buildings in those regions

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Kamakhya Temple, Assam

were renovated, and what survives now are later brick or concrete temples at those sites. It appears that terracotta was the main medium of construction, and also for moulding plaques which depicted Buddhist and Hindu deities in Bengal until the seventh century. A large number of sculptures have been found in Assam and Bengal which shows the development of important regional schools in those regions.

Assam: An old sixth-century sculpted door frame from DaParvatia near Tezpur and another few stray sculptures from Rangagora Tea Estate near Tinsukia in Assam bear witness to the import of the Gupta idiom in that region. This post-Gupta style continued in the region well into the tenth century. However, by the twelfth to fourteenth centuries, a distinct regional style developed in Assam. The style that came with the migration of the Tais from Upper Burma mixed with the dominant Pala style of Bengal and led to the creation of what was later known as the Ahom style in and around Guwahati. Kamakhya temple, *a Shakti Peeth*, is dedicated to Goddess Kamakhya and was built in the seventeenth century.

Bengal: The style of the sculptures during the period between the ninth and eleventh centuries in Bengal (including Bangladesh) and Bihar is known as the Pala style, named after the ruling dynasty at the time, while the style of those of the mid-eleventh to mid-thirteenth centuries is named after the Sena kings. While the Palas are celebrated as patrons of many Buddhist monastic sites, the temples from that region are known to express the local Vanga style. The ninth century Siddheshvara

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Terracotta temple, Vishnupur

Mahadeva temple in Barakar in Burdwan District, for example, shows a tall curving *shikhara* crowned by a large *amalaka* and is an example of the early Pala style. It is similar to contemporaneous temples of Odisha. This basic form grows loftier with the passing of centuries. Many of the temples from the ninth to the twelfth century were located at Telkupi in Purulia District. They were submerged when dams were built in the region. These were amongst the important examples of architectural styles prevalent in the region which showed an awareness of all the known *nagara* sub-types that were prevalent in the rest of North India. However, several temples still survive in Purulia District which can be dated to this period. The black to grey basalt and chlorite stone pillars and arched niches of these temples heavily influenced the earliest Bengal *sultanate* buildings at Gaur and Pandua. Many local vernacular building traditions of Bengal also influenced the style of temples in that region. Most prominent of these was the shape of the curving or sloping side of the bamboo roof of a Bengali hut. This feature was eventually even adopted in Mughal buildings, and is known across North India as the Bangla roof. In the Mughal period and later, scores of terracotta brick temples were built across Bengal and Bangladesh in a unique style that had elements of local building techniques seen in bamboo huts which were combined with older forms reminiscent of the Pala period and with the forms of arches and domes that were taken from Islamic architecture. These can be widely found in and around Vishnupur, Bankura, Burdwan and Birbhum and are dated mostly to the seventeenth century.

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Stone Chariot,
Hampi, Karnataka

Odisha: The main architectural features of Odisha temples are classified in three orders, i.e., *rekhapida*, *pidhadeul* and *khakra*. Most of the main temple sites are located in ancient Kalinga—modern Puri District, including Bhubaneswar or ancient Tribhuvanesvara, Puri and Konark. The temples of Odisha constitute a distinct sub-style within the *nagara* order. In general, here the *shikhara*, called *deul* in Odisha, is vertical almost until the top when it suddenly curves sharply inwards. *Deuls* are preceded, as usual, by *mandapas* called *jagamohana* in Odisha. The ground plan of the main temple is almost always square, which, in the upper reaches of its superstructure becomes circular in the crowning *mastaka*. This makes the spire nearly cylindrical in appearance in its length. Compartments and niches are generally square, the exterior of the temples are lavishly carved, their interiors generally quite bare. Odisha temples usually have boundary walls.

At Konark, on the shores of the Bay of Bengal, lie the majestic ruins of the Surya or Sun temple built in stone around 1240. Its *shikhara* was a colossal creation said to have reached 70m, which, proving too heavy for its site, fell in the nineteenth century. The vast complex is within a quadrilateral precinct of which the *jagamohana* or the dance-pavillion (*mandapa*) has survived, which though no longer accessible is said to be the largest enclosed space in Hindu architecture.

The Sun temple is set on a high base, its walls covered in extensive, detailed ornamental carving. These include twelve pairs of enormous wheels sculpted with spokes and

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Jagannath temple, Puri

hubs, representing the chariot wheels of the Sun god who, in mythology, rides a chariot driven by seven horses, sculpted here at the entrance staircase. The whole temple thus comes to resemble a colossal processional chariot. On the southern wall is a massive sculpture of *surya* carved out of green stone. It is said that there were three such images, each carved out of a different stone placed on the three temple walls, each facing different directions. The fourth wall had the doorway into the temple from where the actual rays of the sun would enter the *garbhagriha*.

The Hills

A unique form of architecture developed in the hills of Kumaon, Garhwal, Himachal and Kashmir. Kashmir's proximity to prominent Gandhara sites (such as Taxila, Peshawar and the northwest frontier) lent the region a strong Gandhara influence by the fifth century CE. This began to mix with the Gupta and post-Gupta traditions that were brought to it from Sarnath, Mathura and even centres in Gujarat and Bengal. Brahmin *pundits* and Buddhist monks frequently travelled between Kashmir, Garhwal, Kumaon and religious centres in the plains like Banaras, Nalanda and even as far south as Kanchipuram. As a result both Buddhist and Hindu traditions began to intermingle and spread in the hills. The hills also had

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Meenakshi temple, Madurai

Of the temples in Kumaon, the ones at Jageshwar near Almora, and Champavat near Pithoragarh, are classic examples of *nagara* architecture in the region.

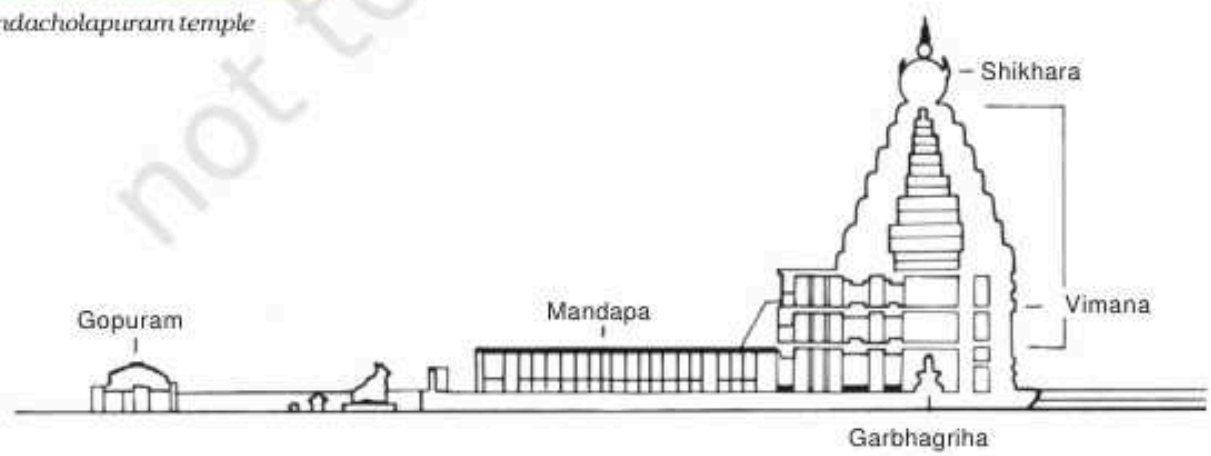
THE *DRAVIDA* OR SOUTH INDIAN TEMPLE STYLE

Unlike *the nagara* temple, the *dravida* temple is enclosed within a compound wall. The front wall has an entrance gateway in its centre, which is known as a *gopuram*. The shape of the main temple tower known as *vimana* in Tamil Nadu is like a stepped pyramid that rises up geometrically rather than the curving *shikhara* of North India. In the South Indian temple, the word '*shikhara*' is used only for the crowning element at the top of the temple which is usually shaped like a small *stupika* or an octagonal cupola—this is equivalent to the *amlak* and *kalasha* of North Indian temples. Whereas at the entrance to the North Indian temple's *garbhagriha*, it would be usual to find images such as *mithunas* and the river goddesses, Ganga and Yamuna, in the south you will generally find sculptures of fierce *dvarapalas* or the door-keepers guarding the temple. It is common to find a large water reservoir, or a temple tank, enclosed within the complex. Subsidiary shrines are either incorporated within the main temple tower, or located as distinct, separate small shrines beside the main temple. The North Indian idea of multiple *shikharas* rising together as a cluster was not popular in South India. At some of the most sacred temples in South India, the main temple in which the *garbhagriha* is situated has, in fact, one of the smallest towers. This is because it is usually the oldest part of the temple. With the passage of time, the population and size of the town associated with that temple would have increased, and it would have become necessary to



Gangaikondacholapuram temple

Dravida temple



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make a new boundary wall around the temple. This would have been taller that the last one, and its *gopurams* would have been even loftier. So, for instance, the Srirangam temple in Tiruchirapally has as many as seven 'concentric' rectangular enclosure walls, each with *gopurams*. The outermost is the newest, while the tower right in the centre housing the *garbhagriha* is the oldest.

Temples thus started becoming the focus of urban architecture. Kanchipuram, Thanjavur or Tanjore, Madurai and Kumbakonam are the most famous temple towns of Tamil Nadu, where, during the eighth to twelfth centuries, the role of the temple was not limited to religious matters alone. Temples became rich administrative centres, controlling vast areas of land.

Just as there are many subdivisions of the main types of *nagara* temples, there are subdivisions also of *dravida* temples. These are basically of five different shapes: square, usually called *kuta*, and also *caturasra*; rectangular or *shala* or *ayatasra*; elliptical, called *gaja-prishta* or elephant-backed, or also called *vrittayata*, deriving from wagon-vaulted shapes of apsidal *chaityas* with a horse-shoe shaped entrance facade usually called a *nasi*; circular or *vritta*; and octagonal or *ashtasra*. Generally speaking, the plan of the temple and the shape of the *vimana* were conditioned by the iconographic nature of the consecrated deity, so it was appropriate to build specific types of temples for specific types of icons. It must, however, be remembered that this is a simplistic differentiation of the subdivisions. Several different shapes may be combined in specific periods and places to create their own unique style.



Shore temple, Mahabalipuram

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Nandi, Brahadeeshwarar

The Pallavas were one of the ancient South Indian dynasties that were active in the Andhra region from the second century CE onwards and moved south to settle in Tamil Nadu. Their history is better documented from the sixth to the eighth century, when they left many inscriptions in stone and several monuments. Their powerful kings spread their empire to various parts of the subcontinent, at times reaching the borders of Odisha, and their links with South–East Asia were also strong. Although they were mostly Shaivite, several Vaishnava shrines also survived from their reign, and there is no doubt that they were influenced by the long Buddhist history of the Deccan.

Their early buildings, it is generally assumed, were rock-cut, while the later ones were structural. However, there is reason to believe that structural buildings were well known even when rock-cut ones were being excavated. The early buildings are generally attributed to the reign of Mahendravarman I, a contemporary of the Chalukyan king, Pulakesin II of Karnataka. Narasimhavarman I, also known as Mamalla, who acceded the Pallava throne around 640 CE, is celebrated for the expansion of the empire, avenging the defeat his father had suffered at the hands of Pulakesin II, and inaugurating most of the building works at Mahabalipuram which is known after him as Mamallapuram.

The shore temple at Mahabalipuram was built later, probably in the reign of Narasimhavarman II, also known as Rajasimha who reigned from 700 to 728 CE. Now it is oriented to the east facing the ocean, but if you study it closely, you will find that it actually houses three shrines,



Brahadeeshwarar, Thanjavur

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Five Rathas, Mahabalipuram

two to Shiva, one facing east and the other west, and a middle one to Vishnu who is shown as *Anantashayana*. This is unusual, because temples generally have a single main shrine and not three areas of worship. This shows that it was probably not originally conceived like this and different shrines may have been added at different times, modified perhaps with the change of patrons. In the compound there is evidence of a water tank, an early example of a *gopuram*, and several other images. Sculptures of the bull, Nandi, Shiva's mount, line the temple walls, and these, along with the carvings on the temple's lower walls have suffered severe disfiguration due to erosion by salt-water laden air over the centuries.

The magnificent Shiva temple of Thanjavur, called the Rajarajeswara or Brahadeeshwarar temple, was completed around 1009 by Rajaraja Chola, and is the largest and tallest of all Indian temples. Temple building was prolific at this time, and over a hundred important temples of the Chola period are in a good state of preservation, and many more are still active shrines. Bigger in scale than anything built by their predecessors, the Pallavas, Chalukyas or Pandyas, this Chola temple's pyramidal multi-storeyed *vimana* rises a massive, 70 metre (230 ft. approx) structure topped by a monolithic *shikhara* which is an octagonal dome-shaped *stupika*. It is in this temple that one notices for the first time two large *gopuras* (gateway towers) with an elaborate sculptural programme which was conceived along with the temple. Huge Nandi-figures dot the corners of the *shikhara*, and the *kalasha* on top by itself is about three metres and eight centimetres in height. Hundreds of stucco figures decorate the *vimana*, although it is possible that some of these may have been added on during the Maratha Period and did not always belong to the Chola Period. The main deity of the temple is Shiva, who is shown as a huge *lingam* set in a two-storeyed

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sanctum. The walls surrounding the sanctum have extended mythological narratives which are depicted through painted murals and sculptures.

Architecture in the Deccan

Many different styles of temple architecture influenced by both North and South Indian temples were used in regions like Karnataka. While some scholars consider the buildings in this region as being distinctly either *nagara* or *dravida*, a hybridised style that seems to have become popular after the mid-seventh century, is known in some ancient texts as *vesara*.

By the late seventh or the early eighth century, the ambitious projects at Ellora became even grander. By about 750 CE, the early western Chalukya control of the Deccan was taken by the Rashtrakutas. Their greatest achievement in architecture is the Kailashnath temple at Ellora, a culmination of at least a millennium-long tradition in rock-cut architecture in India. It is a complete *dravida* building with a Nandi shrine—since the temple is dedicated to Shiva—a *gopuram*-like gateway, surrounding cloisters, subsidiary shrines, staircases and an imposing tower or *vimana* rising to thirty metres. Importantly, all of this is carved out of living rock. One portion of the monolithic hill was carved patiently to build the Kailashnath temple. The sculpture of the Rashtrakuta phase at Ellora is dynamic, the figures often larger than life-size, infused with unparalleled grandeur and the most overwhelming energy.

In the southern part of the Deccan, i.e., in the region of Karnataka is where some of the most experimental



Kailashnath temple, Ellora

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Temple, Badami

hybrid styles of *vesara* architecture are to be found. Pulakesin I established the western Chalukya kingdom when he secured the land around Badami in 543. The early western Chalukyas ruled most of the Deccan till the mid-eighth century when they were superseded by the Rashtrakutas. Early Chalukyan activity also takes the form of rock-cut caves while later activity is of structural temples. The earliest is probably the Ravana Phadi cave at Aihole which is known for its distinctive sculptural style. One of the most important sculptures at the site is of Nataraja, surrounded by larger-than-life-size depictions of the *saptamatrikas*: three to Shiva's left and four to his right. The figures are characterised by graceful, slim bodies, long, oval faces topped with extremely tall cylindrical crowns and shown to wear short *dhotis* marked by fine incised striations indicating pleating. They are distinctly different from



Durga temple, Aihole

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Virupaksha temple, Pattadakal

contemporary western Deccan or Vakataka styles seen at places such as Paunar and Ramtek in Maharashtra.

The hybridisation and incorporation of several styles was the hallmark of Chalukyan buildings. The most elaborate of all Chalukyan temples at Pattadakal made in the reign of Vikramaditya II (733-44) by his chief queen Loka Mahadevi is Virupaksha temple. Another important temple from this site is Papnath temple, dedicated to Lord Shiva. The temple is one of the best early examples of the *Dravida* tradition. By contrast other eastern Chalukyan Temples, like the Mahakuta, five kilometres from Badami, and the Swarga Brahma temple at Alampur show a greater assimilation of northern styles from Odisha and Rajasthan. At the same time the Durga temple at Aihole is unique having an even earlier style of an apsidal shrine which is reminiscent of Buddhist *chaitya* halls and is surrounded by a veranda of a later kind, with a *shikhara* that is stylistically like a *nagara* one. Finally, mention must be made of the Lad Khan temple at Aihole in Karnataka. This seems to be inspired by the wooden-roofed temples of the hills, except that it is constructed out of stone.

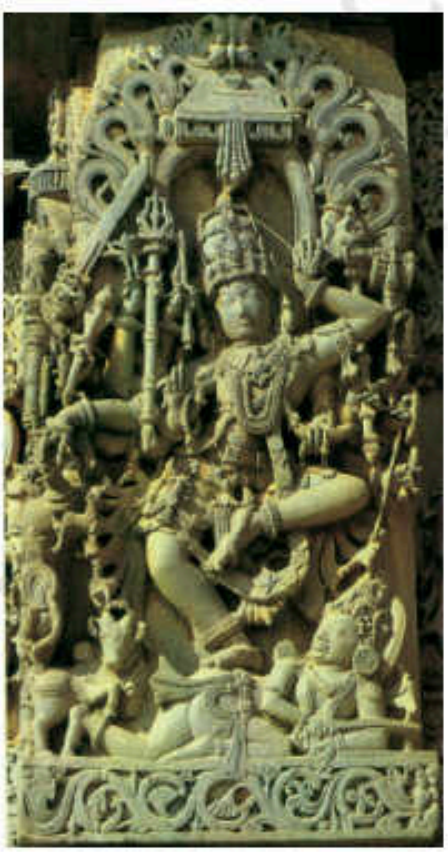
Somnathpuram temple



How then shall we understand these different styles at one place? As curiosities or as innovations? Undoubtedly, they are dynamic expressions of a creative set of architects who were competing with their peers in the rest of India. Whatever one's explanation is, these buildings remain of great art-historical interest.

With the waning of Chola and Pandya power, the Hoysalas of Karnataka grew to prominence in South India and became the most important patrons centred at Mysore. The remains of around hundred temples have been found in southern Deccan, though it is only three of them that are most frequently discussed: the temples at Belur, Halebid and Somnathpuram. Perhaps the most characteristic feature of these temples is that they grow extremely complex with so many projecting angles emerging from the previously straightforward square temple, that the plan of these temples starts looking like a star, and is thus known as a stellate-plan. Since they are made out of soapstone which is a relatively soft stone, the artists were able to carve their sculptures intricately. This can be seen particularly in the jewellery of the gods that adorn their temple walls.

The Hoysaleshvara temple (Lord of the Hoysalas) at Halebid in Karnataka was built in dark schist stone by the Hoysala king in 1150. Hoysala temples are sometimes called hybrid or *vesara* as their unique style seems neither completely *dravida* nor *nagara*, but somewhere in between. They are easily distinguishable from other medieval temples by their highly original star-like ground-plans and a profusion of decorative carvings.



Nataraja, Halebid

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Dedicated to Shiva as Nataraja, the Halebid temple is a double building with a large hall for the *mandapa* to facilitate music and dance. A Nandi pavilion precedes each building. The tower of the temple here and at nearby Belur fell long ago, and an idea of the temples' appearance can now only be gleaned from their detailed miniature versions flanking the entrances. From the central square plan cut-out angular projections create the star effect decorated with the most profuse carvings of animals and deities. So intricate is the carving that it is said, for instance, in the bottom-most frieze featuring a continuous procession of hundreds of elephants with their mahouts, no two elephants are in the same pose.

Founded in 1336, Vijayanagara, literally 'city of victory', attracted a number of international travellers such as the Italian, Niccolo di Conti, the Portuguese Domingo Paes, Fernao Nuniz and Duarte Barbosa and the Afghan Abd al-Razzaq, who have left vivid accounts of the city. In addition, various Sanskrit and Telugu works document the vibrant literary tradition of this kingdom. Architecturally, Vijayanagara synthesises the centuries-old *dravida* temple architecture with Islamic styles demonstrated by the neighbouring sultanates. Their sculpture too, although fundamentally derived from, and consciously seeking to recreate Chola ideals, occasionally shows the presence of foreigners. Their eclectic ruins from the late fifteenth and early sixteenth centuries preserve a fascinating time in history, an age of wealth, exploration and cultural fusion.

BUDDHIST AND JAIN ARCHITECTURAL DEVELOPMENTS

So far, although we have focused on the nature of developments in Hindu architecture from the fifth to fourteenth centuries, it must constantly be kept in mind that this was also the very period when Buddhist and Jain developments were equally vibrant, and often went hand-in-glove with Hindu ones. Sites such as Ellora have Buddhist, Hindu and Jain monuments; however, Badami, Khajuraho and Kannauj have the remains of any two of the religions right next to each other.

When the Gupta empire crumbled in the sixth century CE, this eastern region of Bihar and Bengal, historically known as Magadha, appears to have remained unified whilst numerous small Rajput principalities sprang up to the west. In the eighth century, the Palas came to power in the region. The second Pala ruler, Dharmapala, became immensely powerful and established an empire by defeating

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the powerful Rajput Pratiharas. Dharmapala consolidated an empire whose wealth lay in a combination of agriculture along the fertile Ganges plain and international trade.

The pre-eminent Buddhist site is, of course, Bodhgaya. Bodhgaya is a pilgrimage site since Siddhartha achieved enlightenment here and became Gautama Buddha. While the *bodhi* tree is of immense importance, the Mahabodhi Temple at Bodhgaya is an important reminder of the brickwork of that time. The first shrine here, located at the base of the Bodhi tree, is said to have been constructed by King Ashoka; the *vedika* around it is said to be post-Mauryan, of about 100 BCE; many of the sculptures in the niches in the temple are dated to the eighth century Pala Period, while the actual Mahabodhi temple itself as it stands now is largely a Colonial Period reconstruction of the old seventh century design. The design of the temple is unusual. It is, strictly speaking, neither *dravida* or *nagara*. It is narrow like a *nagara* temple, but it rises without curving, like a *dravida* one.



Mahabodhi temple, Bodhgaya

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7

INDIAN BRONZE SCULPTURE

INDIAN sculptors had mastered the bronze medium and the casting process as much as they had mastered terracotta sculpture and carving in stone. The *cire-perdu* or 'lost-wax' process for casting was learnt as long ago as the Indus Valley Culture. Along with it was discovered the process of making alloy of metals by mixing copper, zinc and tin which is called bronze.

Bronze sculptures and statuettes of Buddhist, Hindu and Jain icons have been discovered from many regions of India dating from the second century until the sixteenth century. Most of these were used for ritual worship and are characterised by exquisite beauty and aesthetic appeal. At the same time the metal-casting process continued to be utilised for making articles for various purposes of daily use, such as utensils for cooking, eating, drinking, etc. Present-day tribal communities also utilise the 'lost-wax' process for their art expressions.

Perhaps the 'Dancing Girl' from Mohenjodaro is the earliest bronze sculpture datable to 2500 BCE. The limbs and torso of this female figurine are simplified in tubular form. A similar group of bronze statuettes have been discovered from archaeological excavation at Daimabad (Maharashtra) datable to 1500 BCE. Significant is the 'Chariot', the wheels of which are represented in simple circular shapes while the driver or human rider has been elongated, and the bulls in the forefront are modelled in sturdy forms.

Interesting images of Jain *tirthankaras* have been discovered from Chausa, Bihar, belonging to the Kushana Period during second century CE. These bronzes show how the Indian sculptors had mastered the modelling of masculine human physique and simplified muscles. Remarkable is the depiction of Adinath or Vrishabhnath, who is identified with long hairlocks dropping to his shoulders. Otherwise the *tirthankaras* are noted by their short curly hair.

Gujarat and Rajasthan have been strongholds of Jainism since early times. A famous hoard of Jain bronzes was found at Akota, on the outskirts of Baroda, dated between



Kaliyadaman, Chola bronze, Tamil Nadu

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the end of the fifth and the end of the seventh century CE. Finely cast through the lost-wax process, these bronzes were often subsequently inlaid with silver and copper to bring out the eyes, crowns and details of the textiles on which the figures were seated. Many famous Jain bronzes from Chausa in Bihar are now kept in the Patna Museum. Many Jain bronzes from Hansi in Haryana and from various sites in Tamil Nadu and Karnataka are kept in various museums in India.

The hoard of bronzes discovered in Akota near Vadodara established that bronze casting was practised in Gujarat or western India between the sixth and ninth centuries. Most of the images represent the Jain *tirthankaras* like Mahavira, Parshvanath or Adinath. A new format was invented in which *tirthankaras* are seated on a throne; they can be single or combined in a group of three or in a group of twenty-four *tirthankaras*. Female images were also cast representing *yakshinis* or *Shasanadevis* of some prominent *tirthankaras*. Stylistically, they were influenced by the features of both the Gupta and the Vakataka period bronzes. Chakreshvari is the *Shasanadevi* of Adinath and Ambika is of Neminath.

Many standing Buddha images with right hand in *abhaya mudra* were cast in North India, particularly Uttar Pradesh and Bihar, during the Gupta and Post-Gupta periods, i.e., between the fifth, sixth and seventh centuries. The *sanghati* or the monk's robe is wrapped to cover the shoulders which turns over the right arm, while the other



Shiva Family, tenth century CE, Bihar



Ganesh, seventh century CE, Kashmir

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end of the drapery is wrapped over the left arm. Eventually the pleats are held by the extended hand of the same arm. The drapery falls and spreads into a wide curve at the level of the ankles. The Buddha's figure is modelled in a subtle manner suggesting, at the same time, the thin quality of the cloth. The whole figure is treated with refinement; there is a certain delicacy in the treatment of the torso. The figure appears youthful and proportionate in comparison with the Kushana style. In the typical bronze from Dhanesar Khera, Uttar Pradesh, the folds of the drapery are treated as in the Mathura style, i.e., in a series of drooping down curves. Sarnath-style bronzes have foldless drapery. The outstanding example is that of the Buddha image at Sultanganj, Bihar, which is quite a monumental bronze figure. The typical refined style of these bronzes is the hallmark of the classical quality.

Vakataka bronze images of the Buddha from Phophnar, Maharashtra, are contemporary with the Gupta period bronzes. They show the influence of the Amaravati style of



Devi, Chola bronze, Tamil Nadu

The Lost-wax Process

The lost-wax process is a technique used for making objects of metal, especially in Himachal Pradesh, Odisha, Bihar, Madhya Pradesh and West Bengal. In each region, a slightly different technique is used.

The lost-wax process involves several different steps. First a wax model of the image is made by hand of pure beeswax that has first been melted over an open fire, and then strained through a fine cloth into a basin of cold water. Here it resolidifies immediately. It is then pressed through a *pichki* or *pharni* — which squeezes the wax into noodle-like shape. These wax wires are then wound around to the shape of the entire image.

The image is now covered with a thick coating of paste, made of equal parts of clay, sand and cow-dung. Into an opening on one side, a clay pot is fixed. In this molten metal is poured. The weight of the metal to be used is ten times that of wax. (The wax is weighed before starting the entire process.) This metal is largely scrap metal from broken pots and pans. While the molten metal is poured in the clay pot, the clay-plastered model is exposed to firing. As the wax inside melts, the metal flows down the channel and takes on the shape of the wax image. The firing process is carried out almost like a religious ritual and all the steps take place in dead silence. The image is later chiselled with files to smoothen it and give it a finish. Casting a bronze image is a painstaking task and demands a high degree of skill. Sometimes an alloy of five metals — gold, silver, copper, brass and lead — is used to cast bronze images.



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Ganesh, Kashmir, seventh century CE

Andhra Pradesh in the third century CE and at the same time there is a significant change in the draping style of the monk's robe. Buddha's right hand in *abhaya mudra* is free so that the drapery clings to the right side of the body contour. The result is a continuous flowing line on this side of the figure. At the level of the ankles of the Buddha figure the drapery makes a conspicuous curvilinear turn, as it is held by the left hand.

The additional importance of the Gupta and Vakataka bronzes is that they were portable and monks carried them from place to place for the purpose of individual worship or to be installed in Buddhist *viharas*. In this manner the refined classical style spread to different parts of India and to Asian countries overseas.

Himachal Pradesh and Kashmir regions also produced bronze images of Buddhist deities as well as Hindu gods and goddesses. Most of these were created during the eighth, ninth and tenth centuries and have a very distinct style in comparison with bronzes from other parts of India. A noteworthy development is the growth of different types of iconography of Vishnu images. Four-headed Vishnu, also known as *Chaturanana* or *Vaikuntha Vishnu*, was worshipped in these regions. While the central face represents Vasudeva,



Bronze sculpture, Himachal Pradesh

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the other two faces are that of Narasimha and Varaha. The Narasimha *avatar* and Mahishasuramardini Durga images of Himachal Pradesh are among the very dynamic bronzes from that region.

In Buddhist centres like Nalanda, a school of bronze-casting emerged around the ninth century during the rule of the Pala Dynasty in Bihar and Bengal regions. In the gap of a few centuries the sculptors at Kurkihar near Nalanda were able to revive the classical style of the Gupta period. A remarkable bronze is of a four-armed Avalokitesvara, which is a good example of a male figure in graceful *tribhanga* posture. Worship of female goddesses was adopted which is part of the growth of the Vajrayana phase in Buddhism. Images of Tara became very popular. Seated on a throne, she is accompanied by a growing curvilinear lotus stalk and her right hand is in the *abhaya mudra*.

The bronze casting technique and making of bronze images of traditional icons reached a high stage of development in South India during the medieval period. Although bronze images were modelled and cast during the Pallava Period in the eighth and ninth centuries, some of the most beautiful and exquisite statues were produced during the Chola Period in Tamil Nadu from the tenth to the twelfth century. The technique and art of fashioning bronze images is still skillfully practised in South India, particularly in Kumbakonam. The distinguished patron



Nataraja, Chola period, twelfth century CE



Nataraja

Shiva is associated with the end of the cosmic world with which this dancing position is associated.

In this Chola period bronze sculpture he has been shown balancing himself on his right leg and suppressing the *apasmara*, the demon of ignorance or forgetfulness, with the foot of the same leg. At the same time he raises his left leg in *bhujangatrasita* stance, which represents *tirobhava*, that is kicking away the veil of *maya* or illusion from the devotee's mind. His four arms are outstretched and the main right hand is posed in *abhaya hasta* or the gesture suggesting. The upper right holds the *damaru* his favourite musical instrument to keep on the beat *tala*. The upper left hand carries a flame while the main left hand is held in *dola hasta* and connects with the *abhaya hasta* of the right hand. His hair locks fly on both the sides touching the circular *jvala mala* or the garland of flames which surrounds the entire dancing figuration.



during the tenth century was the widowed Chola queen, Sembiyan Maha Devi. Chola bronzes are the most sought-after collectors' items by art lovers all over the world.

Among the Pallava Period bronzes of the eighth century is the icon of Shiva seated in *ardhaparyanka asana* (one leg kept dangling). The right hand is in the *achamana mudra* gesture, suggesting that he is about to drink poison.

The well-known dancing figure of Shiva as Nataraja was evolved and fully developed during the Chola Period and since then many variations of this complex bronze image have been modelled.

A wide range of Shiva iconography was evolved in the Thanjavur (Tanjore) region of Tamil Nadu. The ninth century *kalyanasundara murti* is highly remarkable for the manner in which *Panigrahana* (ceremony of marriage) is represented by two separate statuettes. Shiva with his extended right hand accepts Parvati's (the bride's) right hand, who is depicted with a bashful expression and taking a step forward. The union of Shiva and Parvati is very ingeniously represented in the *ardhanarisvara murti* in a single image. Beautiful independent figurines of Parvati have also been modelled, standing in graceful *tribhanga* posture.

During the sixteenth century, known as the Vijayanagar Period in Andhra Pradesh, the sculptors experimented with portrait sculpture in order to preserve knowledge of the royal patron for posterity. At Tirupati, life-size standing portrait statues were cast in bronze, depicting Krishnadevaraya with his two queens, Tirumalamba and Chinnadevi. The sculptor has combined the likeness of the facial features with certain elements of idealisation. The idealisation is further observed in the manner the physical body is modelled to appear imposing as well as graceful. The standing king and queens are depicted in praying posture, that is, both hands held in the *namaskara mudra*.



EXERCISE

1. Do you think that the technique of bronze casting has been a continuous process? How did it evolve over a period of time?
2. In India sculpturing in stone and metal happened simultaneously. In your opinion what were the similarities and differences between both technically, stylistically and functionally?
3. Why are Chola bronze sculptures considered as the most refined?
4. Search for visuals of bronze sculptures of the Buddha belonging to periods other than the Chola Period from Himachal Pradesh, Kashmir, etc.



8

Some Aspects of Indo-Islamic Architecture

In the seventh and eighth centuries CE, Islam spread towards Spain and India. Islam came to India, particularly, with Muslim merchants, traders, holy men and conquerors over a passage of six hundred years. Although by the eighth century CE, Muslims had begun to construct in Sind, Gujarat, etc., it was only in the early thirteenth century that large-scale building activity begun under the Delhi Sultanate, established after the Turkish conquest of northern India.

By the twelfth century, India was already familiar with monumental constructions in grandiose settings. Certain techniques and embellishments were prevalent and popular, such as trabeation (brackets, pillars and lintels) to support a flat roof or a small shallow dome. While arches were shaped in wood and stone, these were unable to bear the weight of the top structure. Now, however, the archuate form of construction was introduced gradually in which arches could support the weight of the domes. Such arches needed to be constructed with *voussoirs* (series of interlocking blocks) and fitted with keystones. The domes, resting on pendatives and squinches enabled spanning of large spaces leaving the interiors free of pillars.

A noteworthy aspect of these migrations and conquests was that Muslims absorbed many features of local cultures and traditions and combined them with their own architectural practices. Thus, in the field of architecture, a mix of many structural techniques, stylised shapes, and surface decorations came about through constant interventions of acceptance, rejection or modification of architectural elements. These architectural entities or categories showcasing multiple styles are known as Indo-Saracenic or Indo-Islamic architecture.

Hindus conceived manifestations of god everywhere in multiple forms as part of their religious faith, whereas, Muslims thought of only one with Muhammad as their Prophet. Hence, Hindus adorned all surfaces with sculptures and paintings. Muslims, forbidden to replicate living forms on any surface, developed their religious art and architecture consisting of the arts of arabesque, geometrical patterns and calligraphy on plaster and stone.

Qutub Minar, Delhi



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TYPOLOGIES OF STRUCTURES

Keeping in mind religious and secular necessities, architectural building like mosques for daily prayers, the Jama Masjids, tombs, *dargahs*, *minars*, *hammams*, formally laid out gardens, madrasas, *sarais* or caravansarais, *Kos minars*, etc., were constructed over a period of time. These were thus additions in the existing types of buildings in the sub-continent.

Architectural edifices in the Indian sub-continent, as elsewhere in the world, were constructed by wealthy people. They were, in descending order, rulers and nobles and their families, merchants, merchant guilds, rural elite and devotees of a cult. In spite of the obvious Saracenic, Persian and Turkish influences, Indo-Islamic structures were heavily influenced by prevailing sensibilities of Indian architectural and decorative forms. A lot depended on the availability of materials, limitations of resources and skills and the sense of aesthetics of the patrons. Although religion and religiosity were very important to people of medieval India, as elsewhere, they borrowed architectural elements liberally.

CATEGORIES OF STYLES

The study of Indo-Islamic architecture is conventionally categorised into the Imperial Style (Delhi Sultanate), the Provincial Style (Mandu, Gujarat, Bengal, and Jaunpur), the Mughal Style (Delhi, Agra, and Lahore) and the Deccani Style (Bijapur, Golconda). These categories help in understanding better the specificities of architectural styles rather than putting them in immutable slots.



Intricate jali work, Amer Fort, Jaipur

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ARCHITECTURAL INFLUENCES

Amongst provincial styles, the architecture of Bengal and Jaunpur is regarded as distinct. Gujarat was said to have a markedly regional character for patrons borrowed elements from regional temple traditions such as *toranas*, lintels in *mihrabs*, carvings of bell and chain motifs, and carved panels depicting trees, for tombs, mosques and *dargahs*. The fifteenth century white marble *dargah* of Shaikh Ahmad Khattu of Sarkhej is a good example of provincial style and it heavily influenced the form and decoration of Mughal tombs.

DECORATIVE FORMS

These forms included designing on plaster through incision or stucco. The designs were either left plain or covered with colours. Motifs were also painted on or carved in stone. These motifs included varieties of flowers, both from the sub-continent and places outside, particularly Iran. The lotus bud fringe was used to great advantage in the inner curves of the arches. Walls were also decorated with cypress, *chinar* and other trees as also with flower vases. Many complex designs of flower motifs decorating the ceilings were also to be found on textiles and carpets. In the fourteenth, fifteenth and sixteenth centuries tiles were also used to surface the walls and the domes. Popular colours were blue, turquoise, green and yellow. Subsequently the techniques of tessellation (mosaic designs) and *pietra dura* were made use of for surface decoration particularly in the dado panels of the walls. At times *lapis lazuli* was used in the interior walls or on canopies. Other decorations included arabesque, calligraphy and high and low relief carving and a profuse use of *jalis*. The high relief carving has a three- dimensional look. The arches were plain and squat and sometimes high and pointed. From the sixteenth century onwards arches were designed with trefoil or multiple foliations. Spandrels of the arches were decorated with medallions or bosses. The roof was a mix of the central dome and other smaller domes, *chatris* and tiny minarets. The central dome was topped with an inverted lotus flower motif and a metal or stone pinnacle.



Dado panel on the wall, Agra



Pietra dura work, Agra

MATERIALS FOR CONSTRUCTION

The walls in all buildings were extremely thick and were largely constructed of rubble masonary, which was easily available. These walls were then cased over with *chunam* or limestone plaster or dressed stone. An amazing range of stones were utilised for construction such as quartzite, sandstone, buff, marble, etc. Polychrome tiles were used to great advantage to finish the walls. From the beginning of the seventeenth century, bricks were also used for construction and these imparted greater flexibility to the structures. In this phase there was more reliance on local materials.

FORTS

Building monumental forts with embattlements was a regular feature in medieval times, often symbolising the seat of power of a king. When such a fort was captured by an attacking army the vanquished ruler either lost his complete power or his sovereignty. This was because he had to accept the suzerainty of the victorious king. Some examples of strong, complex edifices which still exercise the imagination of the visitor are the forts of Chittor, Gwalior, Daulatabad, earlier known as Devgiri and Golconda.

Commanding heights were utilised to great advantage to construct forts. These heights gave a good perspective of the region, strategic advantage for security, unfettered and unhindered space to make residential and official complexes while simultaneously creating a sense of awe in the people. Other complexities woven into such topography were concentric circles

Daulatabad Fort



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Gwalior Fort

of outer walls as in Golconda, so that the enemy had to breach these at all stages before getting in.

Daulatabad had several strategic devices to confound the enemy, such as staggered entrances so that gates could not be opened even with the help of elephants. It also had twin forts, one within the other but at a higher elevation and accessed by a complex defence design arrangement. One wrong turn in the labyrinth or complex pathway could lead to the enemy soldier going in circles or falling to his death several hundred feet below.

The Gwalior Fort was invincible because its steep height made it impossible to scale. It had many habitations and usages. Babur, who did not find much merit in many things he saw in Hindustan, was said to have been overawed at the sight of the Gwalior Fort. Chittorgarh bears the distinction of being the largest fort in Asia and was occupied for the longest length of time as the seat of power. It has many types of buildings including *stambhas* or towers to signify victory and bravery. It was replete with numerous water bodies. Innumerable acts of heroism have been associated with the principal people in the fort, forming the substance of many a legend. An interesting aspect associated with forts is that within the palace complexes stylistic and decorative influences were absorbed most liberally.

MINARS

Another form of *stambha* or tower was the *minar*, a common feature in the sub-continent. Two most striking minars of medieval times are the Qutub Minar in Delhi and the Chand Minar at Daulatabad



Chand Minar, Daulatabad

Fort. The everyday use of the *minar* was for the *azaan* or call to prayer. Its phenomenal height, however, symbolised the might and power of the ruler. The Qutub Minar also came to be associated with the much revered saint of Delhi, Khwaja Qutbuddin Bakhtiyar Kaki.

The Qutub Minar, built in the thirteenth century, is a 234-feet-high tapering tower divided into five storeys. The *minar* is a mix of polygonal and circular shapes. It is largely built of red and buff sandstone with some use of marble in the upper storeys. It is characterised by highly decorated balconies and bands of inscriptions intertwined with foliated designs.

Chand Minar, built in the fifteenth century, is a 210- feet-high tapering tower divided into four storeys. Painted peach now, its façade once boasted of chevron patterning on the encaustic tile work and bold bands of Quranic verses. Although it looked like an Iranian monument, it was the combined handiwork of local architects with those from Delhi and Iran.

TOMBS

Monumental structures over graves of rulers and royalty was a popular feature of medieval India. Some well-known examples of such tombs are those of Ghyasuddin Tughlaq, Humayun, Abdur Rahim Khan-i-Khanan in Delhi, Akbar and Itmaduddaula in Agra. According to Anthony Welch, the idea behind the tomb was eternal paradise as a reward for the true believer on the Day of Judgement. This led to the paradisiacal imagery for tomb



Tomb of Itmaduddaula, Agra

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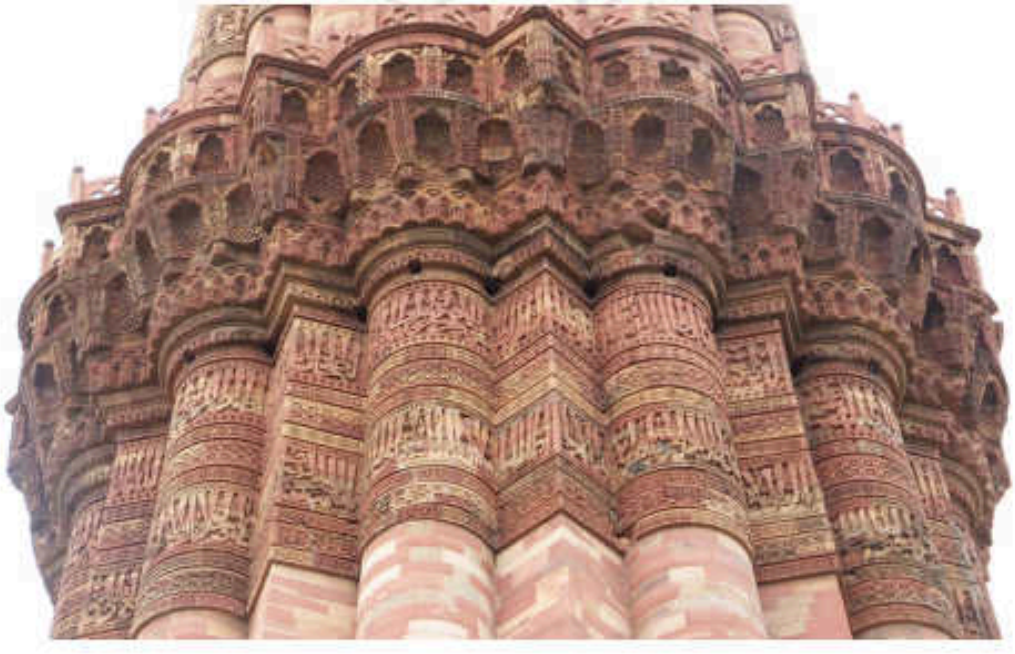
construction. Beginning with the introduction of Quranic verses on the walls, the tomb was, subsequently, placed within paradisiacal elements such as a garden or near a body of water or both, as is in the case of the Humayun's tomb and the Taj Mahal, which follows the *charbagh* style. Surely though, such vast expanses of structured and stylised spaces could not have been intended only to signify peace and happiness in the next world but to also showcase the majesty, grandeur and might of the person buried there.

SARAIS

A hugely interesting feature of medieval India was the *sarais* which ringed cities and dotted the vast space of the Indian sub-continent. *Sarais* were largely built on a simple square or rectangular plan and were meant to provide temporary accommodation to Indian and foreign travellers, pilgrims, merchants, traders, etc. In effect, *sarais* were public domains which thronged with people of varied cultural backgrounds. This led to cross-cultural interaction, influences and syncretic tendencies in the cultural mores of the times and at the level of the people.

STRUCTURES FOR COMMON PEOPLE

One of the architectural features of medieval India was also a coming together of styles, techniques and decorations in public and private spaces of non-royal sections of the society. These included buildings for domestic usage, temples, mosques, *khanqahs* (hermitage of Sufi saints) and *dargahs*, commemorative gateways, pavilions in buildings and gardens, bazaars, etc.





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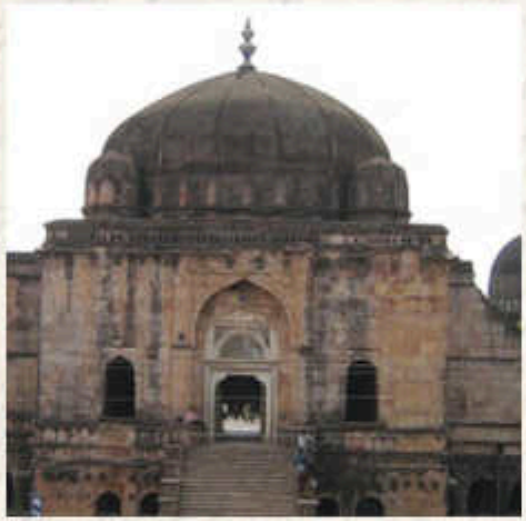


MANDU

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The city of Mandu is located sixty miles from Indore, at an elevation of over 2000 feet and overlooks the Malwa Plateau to the north and the Narmada valley to the south. Mandu's natural defence encouraged consistent habitations by Parmara Rajputs, Afghans and Mughals. As the capital city of Ghauri Dynasty (1401–1561) founded by Hoshang Shah it acquired a lot of fame. Subsequently, Mandu was associated with the romance of Sultan Baz Bahadur and Rani Rupmati. The Mughals resorted to it for pleasure during the monsoon season.

Mandu is a typical respresentation of the medieval provincial style of art and architecture. It was a complex mix of official and residential-cum-pleasure palace, pavilions, mosques, artificial reservoirs, *baolis*, embattlements, etc. In spite of the size or monumentality, the structures were very close to nature, designed in the style of arched pavilions, light and airy, so that these buildings did not retain heat. Local stone and marble were used to great advantage. Mandu is a fine example of architectural adaptation to the environment.



Hoshang Shah's tomb

The Royal Enclave located in the city comprised the most complete and romantic set of buildings, a cluster of palaces and attendant structures, official and residential, built around two artificial lakes. The Hindola Mahal looks like a railway viaduct bridge with its disproportionately large buttresses supporting the walls. This was the audience hall of the Sultan and also the place where he showed himself to his subjects. Batter was used very effectively to give an impression of swinging (*Hindola*) walls.



Hindola Mahal

Jahaaz Mahal is an elegant two-storey 'ship-palace' between two reservoirs, with open pavilions, balconies overhanging the water and a terrace. Built by Sultan Ghiyasuddin Khilji it was possibly used as his harem and as the ultimate pleasure and recreational resort. It had a complex

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Jahaaz Mahal , Mandu

arrangement of watercourses and a terrace swimming pool.

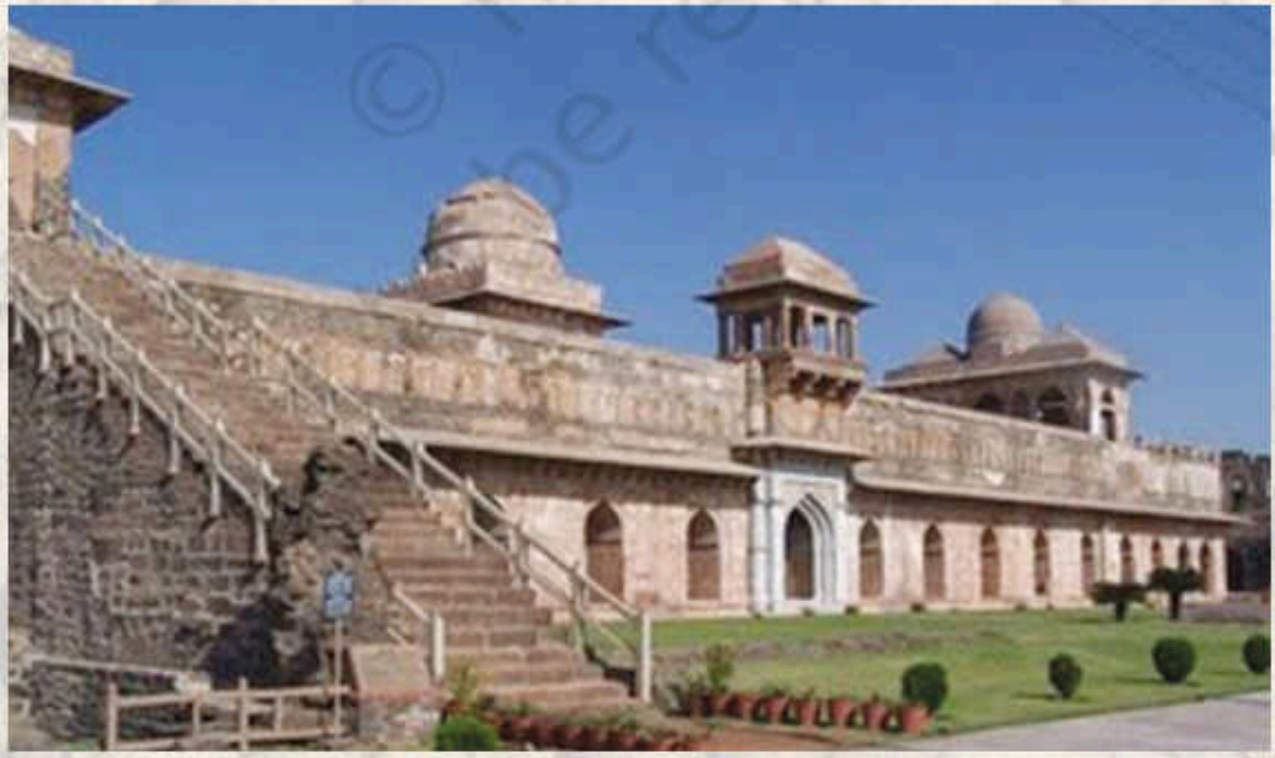
Rani Rupmati's double pavilion perched on the southern embattlements afforded a beautiful view of the Narmada valley. Baz Bahadur's palace had a wide courtyard ringed with halls and terraces.

A *madrasa* called Asharfi Mahal now lies in ruins. Hoshang Shah's tomb is a majestic structure with a beautiful dome, marble *jali* work, porticos, courts and towers. It is regarded as an example of the robustness of Afghan structures, but its lattice work, carved brackets and *toranas* lend it a softer hue.

The *Jama Masjid* of Mandu was built on a large scale to accommodate many worshippers for Friday prayers. It is entered through a monumental gateway, topped with a squat dome, beyond which lies an open courtyard flanked with columned cloisters on three sides, also topped with smaller domes. The building is faced with red sandstone. The *mimbar* in the *Qibla Liwan* is supported on carved brackets and the *mihrab* has a lotus bud fringe.

Provincial style architecture of Mandu is regarded as too close to the structures of Imperial Delhi to make a bold statement of local traditions. Nevertheless, the so-called robust, austere Pathan architecture of Mandu with its surface embellishments of *jalis*, carved brackets, etc., and the lightness of the structures was an important intervention in the narrative of the Indo-Islamic architectural experience.

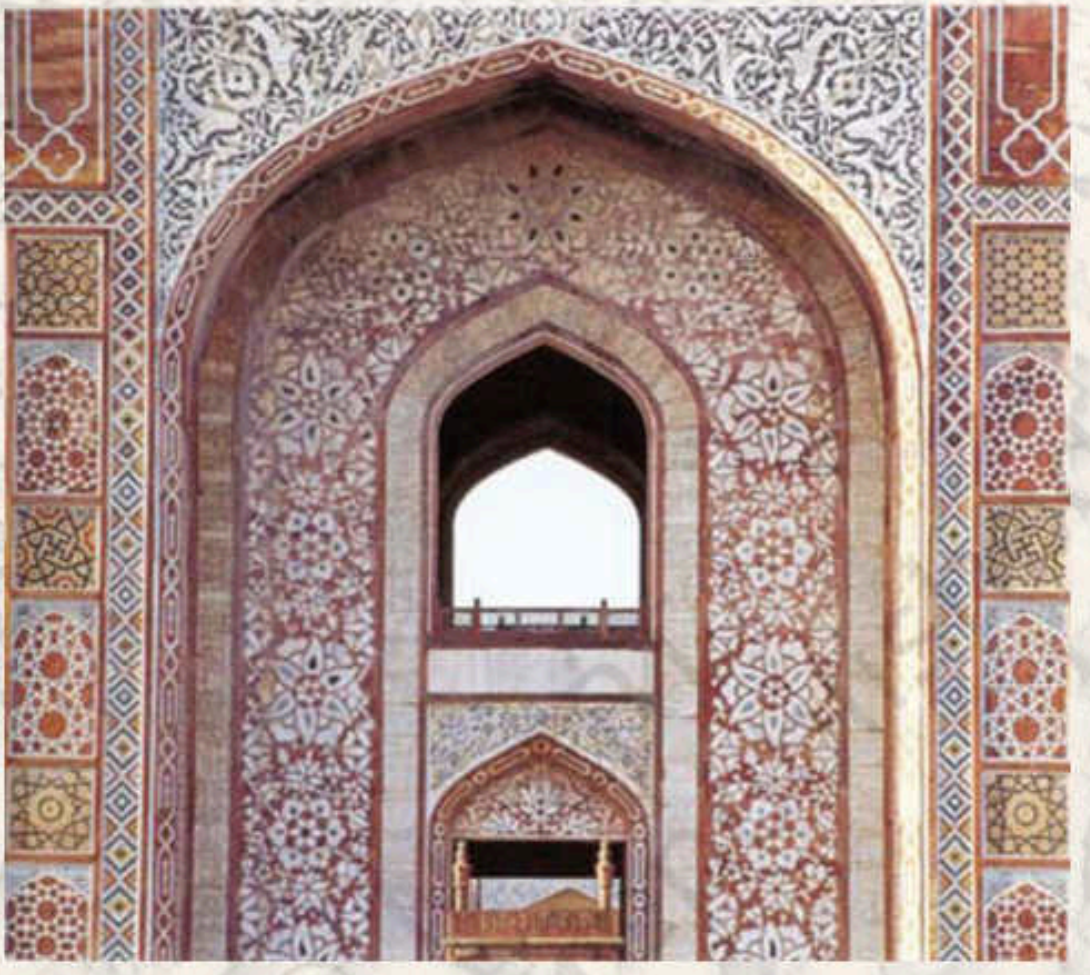
Jama Masjid, Mandu



TAJ MAHAL



From 1632 onwards it took nearly twenty years and 20,000 specialised workers to complete this monument.





Taj Mahal was built in Agra by Shah Jahan as a mausoleum for his deceased wife Mumtaz Mahal. Taj Mahal was the apogee of the evolutionary architectural process in medieval India.

The sublimity of the building comes from its orderly, simple plan and elevation, amazingly perfect proportions or symmetry, the ethereal quality marble has lent to it, the perfect setting of *bagh* and river and the pure outline of the tomb silhouetted against the sky. The patina the Taj has lends it a different hue at various times of day and night.

The Taj complex is entered through a monumental red sandstone gateway the opening arch of which beautifully frames the mausoleum. The tomb is laid out in a Chahar Bagh, criss-crossed with paths and water courses, interspersed with pools and fountains. The structure is placed on the northern extremity of the *bagh* instead of the middle to take advantage of the river bank.

A straight path through the *bagh* reaches the plinth of the tomb from where is accessed the floor terrace of the edifice. At the corners of the terrace stand four tall, tapering minarets, one hundred and thirty-two feet high. The main body of the building is topped with a drum and dome and four cupolas forming a beautiful skyline. The plinth, the walls of the structure and the drum-dome are in perfect proportion to one another. Towards the west of the white marble-faced tomb lies a red sandstone mosque and a similar construction in the east to maintain balance. Marble for the building was quarried from the Makrana mines in Rajasthan and this white edifice is contrasted with the red sandstone of the surrounding structures.

The tomb structure is a square with chamfers forming eight sides, recessed with deep arches. This structural stylisation produces in the elevation of the building a variety of contrasting planes and shade and solids and voids effects. All sides of the building, the twin elevations of floor to roof and roof to pinnacle, atop the foliated crest of the dome, measure 186 feet each.

The interior arrangements of the mausoleum consist of a crypt below and a vaulted, octagonal tomb chamber above, with a room at each angle, all connected with corridors. Light to every part of the building is obtained by means of carved and perforated *jalis*, set in the arched recesses of the interior. The ceiling is as high as the façade creating a void with the help of a double dome.

Four types of embellishments have been used with great effect for the interior and exterior surfaces of the Taj Mahal. These are stone carvings in high and low relief on the walls, the delicate carving of marble into *jalis* and graceful volutes (spiral ornament on the pillar), and the creation of arabesques with pietra dura (yellow marble, jade and jasper) on walls and tombstones and geometric designs with tessellation. Lastly, the art of calligraphy is used with the inlay of jasper in white marble to write Quranic verses. Calligraphy provided a decorative element on the walls and a continuous connection with the Almighty.







Gumbad is situated in Bijapur in the Bijapur District of Karnataka. It is the mausoleum of Muhammad Adil Shah (1626–1656), the seventh Sultan of the Adil Shahi Dynasty of Bijapur (1489–1686). Built by the ruler himself, it is a striking edifice in spite of being unfinished. The tomb is a complex of buildings such as a gateway, a *Naqqar Khana*, a mosque and a *sarai* located within a large-walled garden.

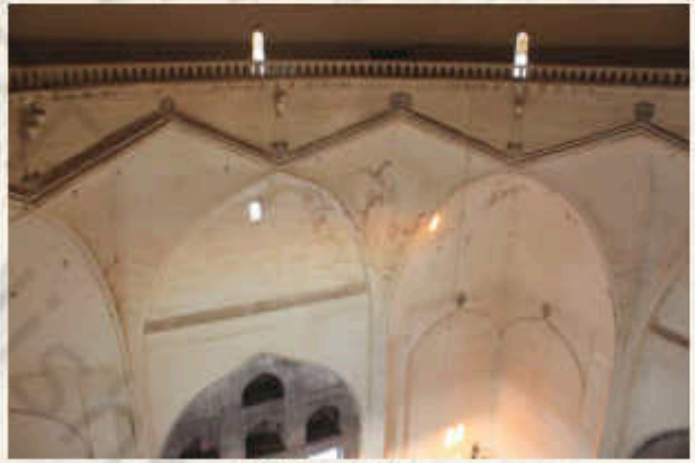


Drum of the Dome

The Gumbad is a monumental square building topped with a circular drum over which rests a majestic dome, giving the building its nomenclature. It is built of dark gray basalt and decorated plasterwork. Each wall of the tomb is one hundred and thirty-five feet long and one hundred and ten feet high and ten feet thick. With the drum and the dome, the building rises to a height of over two hundred feet. The tomb has only one square chamber and the dome, with a diameter of one hundred and twenty-five feet, it covers an uninterrupted floor space of 18,337 square feet, the second largest in the world.

The tomb chamber contains the burial place of the Sultan, his wives and other relatives, while their real graves lie perpendicularly below in a vault, accessed by stairs. The hemispherical masonry dome over a square base was constructed with the help of pendentives. These pendentives not only lent shape to the dome but also transferred its weight to the walls below. New vaulting systems consisting of arch-nets or stellate forms in squinches were created to cover angles formed by intersecting arches.

The building has an amazing acoustical system. Along the drum of the dome there is a whispering gallery where sounds get magnified and echoed many times over.



Whisper Gallery

At the four corners of the building are seven-storeyed octagonal spires or minaret-like towers. These towers house staircases leading to the top dome. The drum of the dome is decorated with foliation. A heavily bracketed cornice resting on corbels is a distinctive feature of the facade.

Gol Gumbad is a fine convergence of many styles located in medieval India. Monumentality, majesty and grandeur, integral aspects of the architectural experience in India, are associated with buildings of Bijapur. While its structural particularities of dome, arches, geometric proportions and load bearing techniques suggest Timurid and Persian styles, it is made of local material and is decorated with surface embellishments popular in the Deccan. Four towers at the corners are reminiscent of turrets attached to mosques such as Qila-i Kuhna Masjid and the Purana Qila in Delhi.

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UNIT I

INTRODUCTION

This unit deals with

- *Location — space relations and India's place in the world*

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CHAPTER

1

India – Location

You have already seen the map of India in the previous classes. Now you closely examine the map of India (Figure 1.1). Mark the southernmost and northernmost latitudes and the easternmost and westernmost longitudes.

The mainland of India, extends from Kashmir in the north to Kanniyakumari in the south and Arunachal Pradesh in the east to Gujarat in the west. India's territorial limit further extends towards the sea upto 12 nautical miles (about 21.9 km) from the coast. (See the box for conversion).

Statute mile	=	63,360 inches
Nautical mile	=	72,960 inches
1 Statute mile	=	about 1.6 km (1.584 km)
1 Nautical mile	=	about 1.8 km (1.852 km)

Our southern boundary extends upto 6°45' N latitude in the Bay of Bengal. Let us try to analyse the implications of having such a vast longitudinal and latitudinal extent.

If you work out the latitudinal and longitudinal extent of India, they are roughly about 30 degrees, whereas the actual distance measured from north to south extremity is 3,214 km, and that from east to west is only 2,933 km. What is the reason for this difference? Consult Chapter 3 on the topic Latitude, Longitude and Time in the book *Practical Work in Geography – Part I* (NCERT, 2006) to find out.

This difference is based on the fact that the distance between two longitudes decreases towards the poles whereas the distance between two latitudes remains the same everywhere. Find out the distance between two latitudes?

From the values of latitude, it is understood that the southern part of the country lies within the tropics and the northern part lies in the sub-tropical zone or the warm temperate zone. This location is responsible for large variations in land forms, climate, soil types and natural vegetation in the country.

Now, let us observe the longitudinal extent and its implications on the Indian people. From the values of longitude, it is quite discernible that there is a variation of nearly 30 degrees, which causes a time difference of nearly two hours between the easternmost and the westernmost parts of our country. You are familiar with the concept of Indian Standard Time (IST). What is the use of the standard meridian? While the sun rises in the northeastern states about two hours earlier as compared to Jaisalmer, the watches in Dibrugarh, Imphal in the east and Jaisalmer, Bhopal or Chennai in the other parts of India show the same time. Why does this happen?

There is a general understanding among the countries of the world to select the standard meridian in multiples of 7°30' of longitude. That is why 82°30' E has been selected as the 'standard meridian' of India. Indian Standard Time is ahead of Greenwich Mean Time by 5 hours and 30 minutes.

There are some countries where there are more than one standard meridian due to their vast east-to-west extent. For example, the USA has seven time zones.



Figure 1.1 : India : Administrative Divisions

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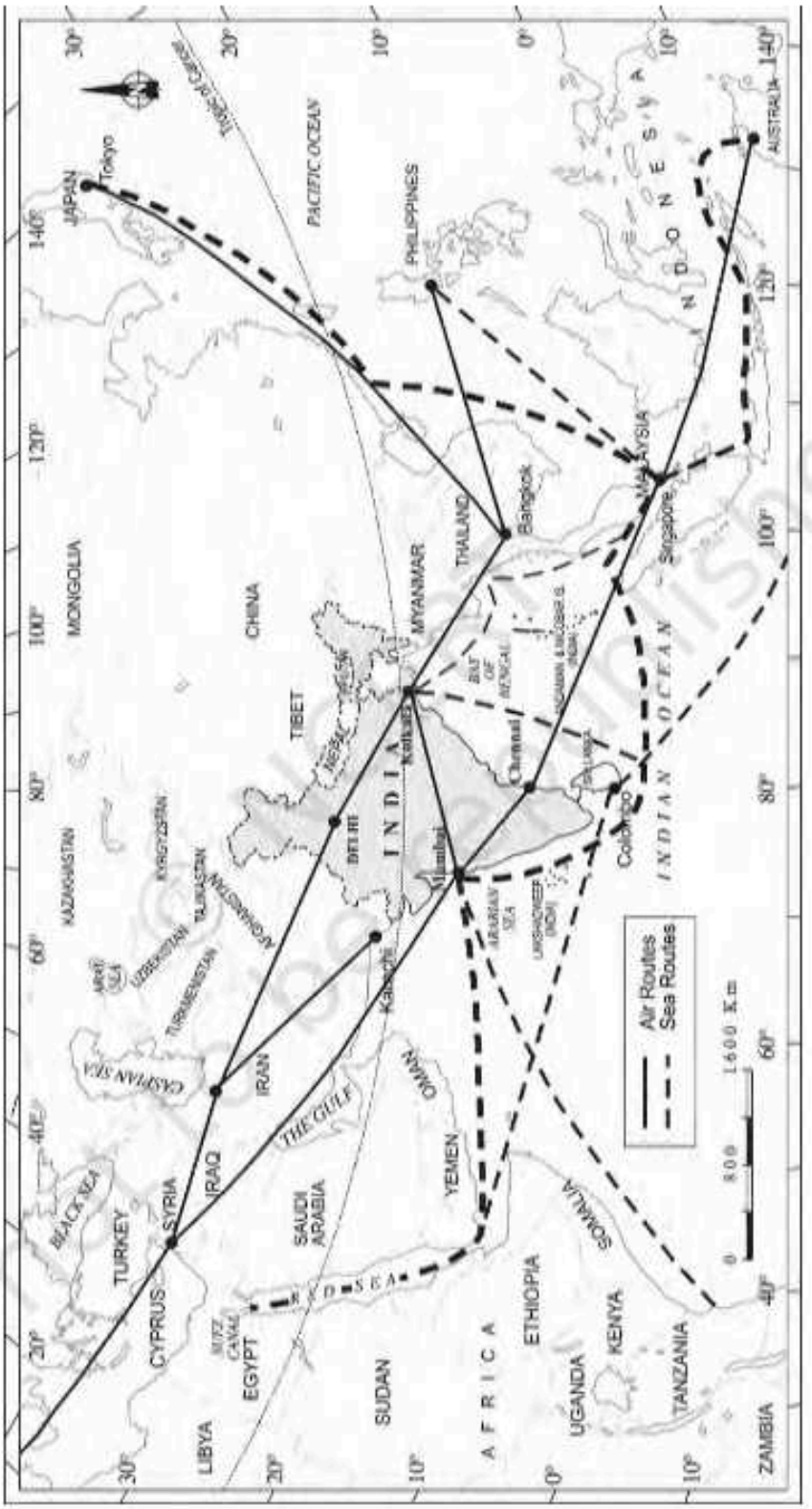


Figure 1.2 : Location of India in the Eastern World

Name a few places in India through which the standard meridian passes?

India with its area of 3.28 million sq. km accounts for 2.4 per cent of the world's land surface area and stands as the seventh largest country in the world. Find out the names of the countries which are larger than India.

Size

The size of India has endowed her with great physical diversity. Thus, you may appreciate the presence of lofty mountains in the north; large rivers such as Ganga, Brahmaputra, Mahanadi, Krishna, Godavari and Kaveri; green forested hills in northeast and south India; and the vast sandy expanse of *Marusthali*. You may further appreciate that bounded by the Himalayas in the north, Hindukush and Sulaiman ranges in the north-west, Purvachal hills in the north-east and by the large expanse of the Indian ocean in the south, it forms a great geographic entity known as the *Indian subcontinent*. It includes the countries — Pakistan, Nepal, Bhutan, Bangladesh and India. The Himalayas, together with other ranges, have acted as a formidable physical barrier in the past. Except for a few mountain passes such as the Khyber, the Bolan, the Shipkila, the Nathula, the Bomdila, etc. it was difficult to cross it. It has contributed towards the evolving of a unique regional identity of the Indian subcontinent.

By referring to the physical map of India you can now describe the physical variations which you would come across while travelling from Kashmir to Kanniyakumari and from Jaisalmer in Rajasthan to Imphal in Manipur.

Peninsular part of India extends towards the Indian Ocean. This has provided the country with a coastline of 6,100 km in the mainland and 7,517 km in the entire geographical coast of the mainland plus the island groups Andaman and Nicobar located in the Bay of Bengal and the Lakshadweep in the Arabian Sea. Thus India, as a country, is a physically diverse land providing occurrence of varied resources.

Do you Remember?

School Bhuvan NCERT is a portal providing map-based learning to bring awareness among the students about country's natural resources, environment and their role in sustainable development. It is an initiative of Bhuvan-NRSC/ISRO, based on NCERT syllabus. You can explore various maps of India on http://bhuvan-app1.nrsc.gov.in/mhrd_ncert/

India and its Neighbours

Examine the location map of India (Figure 1.2). You will notice that India is located in the south-central part of the continent of Asia, bordering the Indian ocean and its two arms extending in the form of Bay of Bengal and the Arabian Sea. This maritime location of Peninsular India has provided links to its neighbouring regions through the sea and air routes.

Prepare a list of India's neighbouring countries by consulting the map.

Sri Lanka and Maldives are the two island countries located in the Indian Ocean, which are our neighbours. Sri Lanka is separated from India by the Gulf of Mannar and Palk Strait.

Differentiate between a Gulf and a Strait.

Do you think that physical barrier is a hindrance in interaction with our neighbouring countries in modern times? Give some examples how we have overcome these difficulties in the present day.

Activity: Observe the map of India on an atlas/School Bhuvan NCERT portal and collect information about states/districts/villages located near the international border of India.

EXERCISES

1. Choose the right answer from the four alternatives given below.

(i) Which one of the following latitudinal extent is relevant for the extent of India's area?

(a) 8°41'N - 35°7'N (c) 8°4'N - 35°6'N
(b) 8°4'N - 37°6'N (d) 6°45'N - 37°6'N

(ii) Which one of the following countries shares the longest land frontier with India?

(a) Bangladesh (c) Pakistan
(b) China (d) Myanmar

(iii) Which one of the following countries is larger in area than India?

(a) China (c) France
(b) Egypt (d) Iran

(iv) Which one of the following longitudes is the standard meridian for India?

(a) 69°30'E (c) 75°30'E
(b) 82°30'E (d) 90°30'E

2. Answer the following questions in about 30 words.

(i) Does India need to have more than one standard time? If yes, why do you think so?

(ii) What are the implications of India having a long coastline?

(iii) How is the latitudinal spread of India advantageous to her?

(iv) While the sun rises earlier in the east, say Nagaland and also sets earlier, how do the watches at Kohima and New Delhi show the same time?

Project/Activity

Activity based on Map of India/Atlas/Census (Teachers may help in the exercises by explaining and getting it done by the students).

(i) On a graph paper, plot the number of districts in Madhya Pradesh, Karnataka, Meghalaya, Goa, Kerala, Haryana. Do the number of districts have some relationship with the area of the state?

(ii) Which state amongst Uttar Pradesh, West Bengal, Gujarat, Arunachal Pradesh, Tamil Nadu, Tripura and Rajasthan is the most thickly populated and which one is the least densely populated?

(iii) Identify the states with coastal boundaries.

(iv) Arrange the states from west to east which have only land boundary.

(v) List the Union Territories which have coastal location.

(vi) How do you explain the variation in the area and population of NCT Delhi and the Andaman and Nicobar Islands?

(vii) On a graph paper, draw a bar diagram to show the area and population of all the Union Territories.

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CHAPTER

2

STRUCTURE AND PHYSIOGRAPHY

Do you know that our earth also has a history. The earth and its landforms that we see today have evolved over a very long time. Current estimation shows that the earth is approximately 4600 million years old. Over these long years, it has undergone many changes brought about primarily by the endogenic and exogenic forces. These forces have played a significant role in giving shape to various surface and subsurface features of the earth. You have already studied about the Plate Tectonics and the movement of the Earth's plates in the book *Fundamentals of Physical Geography* (NCERT, 2024). Do you know that the Indian plate was to the south of the equator millions of years ago? Do you also know that it was much larger in size and the Australian plate was a part of it? Over millions of years, this plate broke into many parts and the Australian plate moved towards the southeastern direction and the Indian plate to the north. Can you map different phases in the movement of the Indian plate? This northward movement of the Indian plate is still continuing and it has significant consequences on the physical environment of the Indian subcontinent. Can you name some important consequences of the northward movement of the Indian plate?

It is primarily through the interplay of these endogenic and exogenic forces and lateral movements of the plates that the present geological structure and geomorphologic processes active in the Indian subcontinent came into existence. Based on the variations in its geological structure and formations, India can be divided into three geological divisions. These geological regions broadly follow the physical features:

(i) The Penisular Block
(ii) The Himalayas and other Peninuslar Mountains
(iii) Indo-Ganga-Brahmaputra Plain.

THE PENINSULAR BLOCK

The northern boundary of the Peninsular Block may be taken as an irregular line running from Kachchh along the western flank of the Aravali Range near Delhi and then roughly parallel to the Yamuna and the Ganga as far as the Rajmahal Hills and the Ganga delta. Apart from these, the Karbi Anglong and the Meghalaya Plateau in the northeast and Rajasthan in the west are also extensions of this block. The northeastern parts are separated by the Malda fault in West Bengal from the Chotanagpur plateau. In Rajasthan, the desert and other desert–like features overlay this block.

The Peninsula is formed essentially by a great complex of very ancient gneisses and granites, which constitutes a major part of it. Since the Cambrian period, the Peninsula has been standing like a rigid block with the exception of some of its western coast which is submerged beneath the sea and some other parts changed due to tectonic activity without affecting the original basement. As a part of the Indo-Australian Plate, it has been subjected to various vertical movements and block faulting. The rift valleys of the Narmada, the Tapi and the Mahanadi and the Satpura block

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mountains are some examples of it. The Peninsula mostly consists of relict and residual mountains like the Aravali hills, the Nallamala hills, the Javadi hills, the Veliconda hills, the Palkonda range and the Mahendragiri hills, etc. The river valleys here are shallow with low gradients.

You are aware of the method of calculating the gradient as a part of your study of the book *Practical Work in Geography– Part I* (NCERT, 2025). Can you calculate the gradient of the Himalayan and the Peninsular rivers and draw the comparisons?

Most of the east flowing rivers form deltas before entering into the Bay of Bengal. The deltas formed by the Mahanadi, the Krishna, the Kaveri and the Godavari are important examples.

The Himalayas and other Peninsular Mountains

The Himalayas along with other Peninsular mountains are young, weak and flexible in their geological structure unlike the rigid and stable Peninsular Block. Consequently, they are still subjected to the interplay of exogenic and endogenic forces, resulting in the development of faults, folds and thrust plains. These



Figure 2.1 : A Gorge

mountains are tectonic in origin, dissected by fast-flowing rivers which are in their youthful stage. Various landforms like gorges, V-shaped valleys, rapids, waterfalls, etc. are indicative of this stage.

Indo-Ganga-Brahmaputra Plain

The third geological division of India comprises the plains formed by the river Indus, the Ganga and the Brahmaputra. Originally, it was a geo-synclinal depression which attained its maximum development during the third phase of the Himalayan mountain formation approximately about 64 million years ago. Since then, it has been gradually filled by the sediments brought by the Himalayan and Peninsular rivers. Average depth of alluvial deposits in these plains ranges from 1,000-2,000 m.

It is evident from the above discussion that there are significant variations among the different regions of India in terms of their geological structure, which has far-reaching impact upon other related aspects. Variations in the physiography and relief are important among these. The relief and physiography of India has been greatly influenced by the geological and geomorphological processes active in the Indian subcontinent.

Physiography

'Physiography' of an area is the outcome of structure, process and the stage of development. The land of India is characterised by great diversity in its physical features. The north has a vast expanse of rugged topography consisting of a series of mountain ranges with varied peaks, beautiful valleys and deep gorges. The south consists of stable table land with highly dissected plateaus, denuded rocks and developed series of scarps. In between these two lies the vast north Indian plain.

Based on these macro variations, India can be divided into the following physiographic divisions:

(1) The Northern and North-eastern Mountains
(2) The Northern Plain

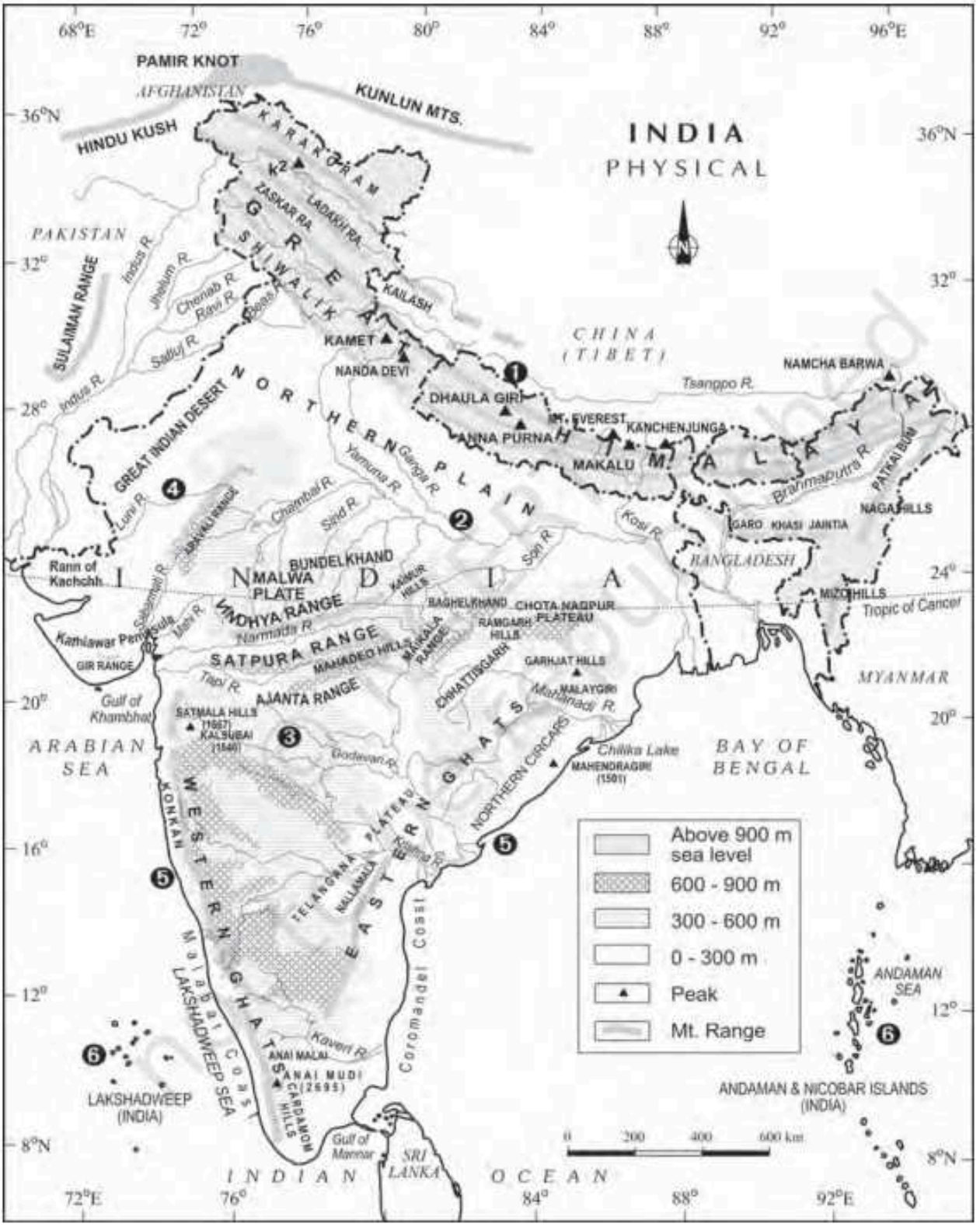


Figure 2.2 : India : Physical

(3) The Peninsular Plateau
(4) The Indian Desert
(5) The Coastal Plains
(6) The Islands.

The North and Northeastern Mountains

The North and Northeastern Mountains consist of the Himalayas and the Northeastern hills. The Himalayas consist of a series of parallel mountain ranges. Some of the important ranges are the Greater Himalayan range, which includes the Great Himalayas and the Shiwalik. The general orientation of these ranges is from northwest to the southeast direction in the northwestern part of India. Himalayas in the Darjiling and Sikkim regions lie in an eastwest direction, while in Arunachal Pradesh they are from southwest to the northwest direction. In Nagaland, Manipur and Mizoram, they are in the northsouth direction. The approximate length of the Great Himalayan range, also known as the central axial range, is 2,500 km from east to west, and their width varies between 160-400 km from north to south. It is also evident from the map that the Himalayas stand almost like a strong and long wall between the Indian subcontinent and the Central and East Asian countries.

Himalayas are not only the physical barrier, they are also a climatic, drainage and cultural divide. Can you identify the impact of Himalayas on the geoenvironment of the countries of South Asia? Can you find some other examples of similar geoenvironmental divide in the world?



Figure 2.3 : The Himalayas

The Northern Plains

The northern plains are formed by the alluvial deposits brought by the rivers – the Indus, the Ganga and the Brahmaputra. These plains extend approximately 3,200 km from the east to the west. The average width of these plains varies between 150-300 km. The maximum depth of alluvium deposits varies between 1,000-2,000 m. From the north to the south, these can be divided into three major zones: the *Bhabar*, the *Tarai* and the alluvial plains. The alluvial plains can be further divided into the *Khadar* and the *Bhangar*.

Bhabar is a narrow belt ranging between 8-10 km parallel to the Shiwalik foothills at the break-up of the slope. As a result of this, the streams and rivers coming from the mountains deposit heavy materials of rocks and boulders, and at times, disappear in this zone. South of the *Bhabar* is the *Tarai* belt, with an approximate width of 10-20 km where most of the streams and rivers re-emerge without having any properly demarcated channel, thereby, creating marshy and swampy conditions known as the *Tarai*. This has a luxurious growth of natural vegetation and houses a varied wildlife.

The south of *Tarai* is a belt consisting of old and new alluvial deposits known as the *Bhangar* and *Khadar* respectively. These plains have characteristic features of mature stage of fluvial erosional and depositional landforms such as sand bars, meanders, ox-bow lakes and braided channels. The Brahmaputra plains are known for their riverine islands and sand bars. Most of these areas are subjected to periodic floods



Figure 2.4 : Northern Plain

and shifting river courses forming braided streams.

The mouths of these mighty rivers also form some of the largest deltas of the world, for example, the famous Sunderbans delta. Otherwise, this is a featureless plain with a general elevation of 50-150 m above the mean sea level. The states of Haryana and Delhi form a water divide between the Indus and the Ganga river systems. As opposed to this, the Brahmaputra river flows from the northeast to the southwest direction before it takes an almost 90° southward turn at Dhubri before it enters into Bangladesh. These river valley plains have a fertile alluvial soil cover which supports a variety of crops like wheat, rice, sugarcane and jute, and hence, supports a large population.

The Peninsular Plateau

Rising from the height of 150 m above the river plains up to an elevation of 600-900 m is the irregular triangle known as the Peninsular plateau. Delhi ridge in the northwest, (extension of Aravalis), the Rajmahal hills in the east, Gir range in the west and the Cardamom hills in the south constitute the outer extent of the Peninsular plateau. However, an extension of this is also seen in the northeast, in the form of Shillong and Karbi-Anglong plateau. The Peninsular India is made up of a series of patland plateaus such as the Hazaribagh plateau, the Palamu plateau, the Ranchi plateau, the Malwa plateau, the Coimbatore plateau and the Karnataka plateau, etc. This is one of the oldest and the most stable landmass of India. The general elevation of the plateau is from the west to the east, which is also proved by the pattern of the flow of rivers. Name some rivers of the Peninsular plateau which have their confluence in the Bay of Bengal and the Arabian sea and mention some landforms which are typical to the east flowing rivers but are absent in the west flowing rivers. Some of the important physiographic features of this region are tors, block mountains, rift valleys, spurs, bare rocky structures, series of hummocky hills and wall-like quartzite dykes offering natural sites for water storage. The western and northwestern part of the plateau has an emphatic presence of black soil.



Figure 2.5 : A Part of Peninsular Plateau

This Peninsular plateau has undergone recurrent phases of upliftment and submergence accompanied by crustal faulting and fractures. (The Bhima fault needs special mention, because of its recurrent seismic activities). These spatial variations have brought in elements of diversity in the relief of the Peninsular plateau. The northwestern part of the plateau has a complex relief of ravines and gorges. The ravines of Chambal, Bhind and Morena are some of the well-known examples.

On the basis of the prominent relief features, the Peninsular plateau can be divided into three broad groups:

(i) The Deccan Plateau
(ii) The Central Highlands
(iii) The Northeastern Plateau.

The Deccan Plateau

This is bordered by the Western Ghats in the west, Eastern Ghats in the east and the Satpura, Maikal range and Mahadeo hills in the north. Western Ghats are locally known by different names such as Sahyadri in Maharashtra, Nilgiri hills in Karnataka and Tamil Nadu and Anaimalai hills and Cardamom hills in Kerala. Western Ghats are

comparatively higher in elevation and more continuous than the Eastern Ghats. Their average elevation is about 1,500 m with the height increasing from north to south. 'Anaimudi' (2,695 m), the highest peak of Peninsular plateau is located on the Anaimalai hills of the Western Ghats followed by Dodabetta (2,637 m) on the Nilgiri hills. Most of the Peninsular rivers have their origin in the Western Ghats. Eastern Ghats comprising the discontinuous and low hills are highly eroded by the rivers such as the Mahanadi, the Godavari, the Krishna, the Kaveri, etc. Some of the important ranges include the Javadi hills, the Palconda range, the Nallamala hills, the Mahendragiri hills, etc. The Eastern and the Western Ghats meet each other at the Nilgiri hills.

The Central Highlands

They are bounded to the west by the Aravali range. The Satpura range is formed by a series of scarped plateaus on the south, generally at an elevation varying between 600-900 m above the mean sea level. This forms the northernmost boundary of the Deccan plateau. It is a classic example of the relict mountains which are highly denuded and form discontinuous ranges. The extension of the Peninsular plateau can be seen as far as Jaisalmer in the West, where it has been covered by the longitudinal sand ridges and crescent-shaped sand dunes called *barchans*. This region has undergone metamorphic processes in its geological history, which can be corroborated by the presence of metamorphic rocks such as marble, slate, gneiss, etc.

The general elevation of the Central Highlands ranges between 700-1,000 m above the mean sea level and it slopes towards the north and northeastern directions. Most of the tributaries of the river Yamuna have their origin in the Vindhyan and Kaimur ranges. Banas is the only significant tributary of the river Chambal that originates from the Aravalli in the west. An eastern extension of the Central Highland is formed by the Rajmahal hills, to the south of which lies a large reserve of mineral resources in the Chotanagpur plateau.

The Northeastern Plateau

In fact it is an extension of the main Peninsular plateau. It is believed that due to the force exerted by the northeastward movement of the Indian plate at the time of the Himalayan origin, a huge fault was created between the Rajmahal hills and the Meghalaya plateau. Later, this depression got filled up by the deposition activity of the numerous rivers. Today, the Meghalaya and Karbi Anglong plateau stand detached from the main Peninsular Block. The Meghalaya plateau is further sub-divided into three: (i) The Garo Hills; (ii) The Khasi Hills; (iii) The Jaintia Hills, named after the tribal groups inhabiting this region. An extension of this is also seen in the Karbi Anglong hills of Assam. Similar to the Chotanagpur plateau, the Meghalaya plateau is also rich in mineral resources like coal, iron ore, sillimanite, limestone and uranium. This area receives maximum rainfall from the south west monsoon. As a result, the Meghalaya plateau has a highly eroded surface. Cherrapunji displays a bare rocky surface devoid of any permanent vegetation cover.

The Indian Desert

To the northwest of the Aravali hills lies the Great Indian desert. It is a land of undulating topography dotted with longitudinal dunes and *barchans*. This region receives low rainfall below 150 mm per year; hence, it has arid climate with low vegetation cover. It is because of these characteristic features that this is also known as *Marusthali*. It is believed that during the Mesozoic era, this region was under the sea. This can be corroborated by the evidence available at wood fossils park at Aakal and marine deposits around Brahmsar, near Jaisalmer (The approximate age of the wood-fossils is estimated to be 180 million years). Though the underlying rock structure of the desert is an extension of the Peninsular plateau, yet, due to extreme arid conditions, its surface



Figure 2.6 : The Indian Desert

Can you identify the type of sand dunes shown in this picture?

features have been carved by physical weathering and wind actions. Some of the well pronounced desert land features present here are mushroom rocks, shifting dunes and oasis (mostly in its southern part). On the basis of the orientation, the desert can be divided into two parts: the northern part is sloping towards Sindh and the southern towards the Rann of Kachchh. Most of the rivers in this region are ephemeral. The Luni river flowing in the southern part of the desert is of some significance. Low precipitation and high evaporation makes it a water deficit region. There are some streams which disappear after flowing for some distance and present a typical case of inland drainage by joining a lake or playa. The lakes and the playas have brackish water which is the main source of obtaining salt.

The Coastal Plains

You have already read that India has a long coastline . On the basis of the location and active geomorphological processes, it can be broadly divided into two: (i) the western coastal plains; (ii) the eastern coastal plains.

The western coastal plains are an example of submerged coastal plain. It is believed that the city of Dwaraka which was once a part of the Indian mainland situated along the west coast is submerged under water. Because of this submergence it is a narrow belt and provides natural conditions for the development of ports and harbours. Kandla, Mazagaon, JLN port Navha Sheva, Marmagao, Mangalore, Cochin, etc. are some of the important natural ports located along the west coast. Extending from the Gujarat coast in the north to the Kerala coast in the south, the western coast may be divided into following divisions – the Kachchh and Kathiawar coast in Gujarat, Konkan coast in Maharashtra, Goan coast and Malabar coast in Karnataka and Kerala respectively. The western coastal plains are narrow in the middle and get broader towards north and south. The rivers flowing through this coastal plain do not form any delta. The Malabar coast has got certain distinguishing features in the form of *'Kayals'* (backwaters), which are used for fishing, inland navigation and also due to its special attraction for tourists. Every year the famous *Nehru Trophy Vallamkali* (boat race) is held in *Punnamada Kayal* in Kerala.

As compared to the western coastal plain, the eastern coastal plain is broader and is an example of an emergent coast. There are well-developed deltas here, formed by the rivers flowing eastward in to the Bay of Bengal. These include the deltas of the Mahanadi, the Godavari, the Krishna and the Kaveri. Because of its emergent nature, it has less number of ports and harbours. The continental shelf extends up to 500 km into the sea, which makes it difficult for the development of good ports and harbours. Name some ports on the eastern coast.



Figure 2.7 : Coastal Plains

On 26 December 2004, the Andaman and Nicobar Islands experienced one of the most devasting natural calamity. Can you name the calamity and identify some other areas which were adversely affected by the same calamity? What was its major consequence?

The Islands

There are two major island groups in India – one in the Bay of Bengal and the other in the Arabian Sea. The Bay of Bengal island groups consist of about 572 islands/islets. These are situated roughly between 6°°N-14°°N and 92°°E -94°°E. The two principal groups of islets include the Ritchie's archipelago and the Labrynth island. The entire group of island is divided into two broad categories – the Andaman in the north and the Nicobar in the south. They are separated by a waterbody which is called the Ten degree channel. It is believed that these islands are an elevated portion of submarine mountains. However, some smaller islands are volcanic in origin. *Barren island*, the only active volcano in India is also situated in the Andaman and Nicobar islands.

Some important mountain peaks in Andaman and Nicobar Islands are Saddle peak (North Andaman – 738 m), Mount Diavolo (Middle Andaman – 515 m), Mount Koyob (South Andaman – 460 m) and Mount Thuiller (Great Nicobar – 642 m).

The coastal line has some coral deposits, and beautiful beaches. These islands receive convectional rainfall and have an equatorial type of vegetation.

The islands of the Arabian sea include Lakshadweep and Minicoy. These are scattered between 8°°N-12°°N and 71°°E-74°°E longitude. These islands are located at a distance of 220 km-440 km off the Kerala coast. The entire island group is built of coral deposits. There are approximately 36 islands of which 11 are inhabited. *Minicoy* is the largest island with an area of 453 sq. km. The entire group of islands is broadly divided by the Nine degree channel, north of which is the Amini Island and to the south of the Canannore Island. The Islands of this archipelago have storm beaches consisting of unconsolidated pebbles, shingles, cobbles and boulders on the eastern seaboard.



Figure 2.8 : An Island

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CHAPTER

3

DRAINAGE SYSTEM

You have observed water flowing through the rivers, *nalas* and even channels during rainy season which drain the excess water. Had these channels not been there, large-scale flooding would have occurred. Wherever channels are ill-defined or choked, flooding is a common phenomenon.

> The flow of water through well-defined channels is known as 'drainage' and the network of such channels is called a 'drainage system'. The drainage pattern of an area is the outcome of the geological time period, nature and structure of rocks, topography, slope, amount of water flowing and the periodicity of the flow.

Do you have a river near your village or city? Have you ever been there for boating or bathing? Is it perennial (always with water) or ephemeral (water during rainy season, and dry, otherwise)? Do you know that rivers flow in the same direction? You have studied about slopes in the other two textbooks of geography (NCERT, 2024) in this class . Can you, then, explain the reason for water flowing from one direction to the other? Why do the rivers originating from the Himalayas in the northern India and the Western Ghat in the southern India flow towards the east and discharge their waters in the Bay of Bengal?



Figure 3.1 : A River in the Mountainous Region

A river drains the water collected from a specific area, which is called its 'catchment area'.

An area drained by a river and its tributaries

Important Drainage Patterns

(i) The drainage pattern resembling the branches of a tree is known as "dendritic" the examples of which are the rivers of northern plain.

(ii) When the rivers originate from a hill and flow in all directions, the drainage pattern is known as 'radial'. The rivers originating from the Amarkantak range present a good example of it.

(iii) When the primary tributaries of rivers flow parallel to each other and secondary tributaries join them at right angles, the pattern is known as 'trellis'.

(iv) When the rivers discharge their waters from all directions in a lake or depression, the pattern is know as 'centripetal'.

Find out some of the patterns in the topo sheet given in Chapter 5 of *Practical Work in Geography– Part 1 (NCERT)*.

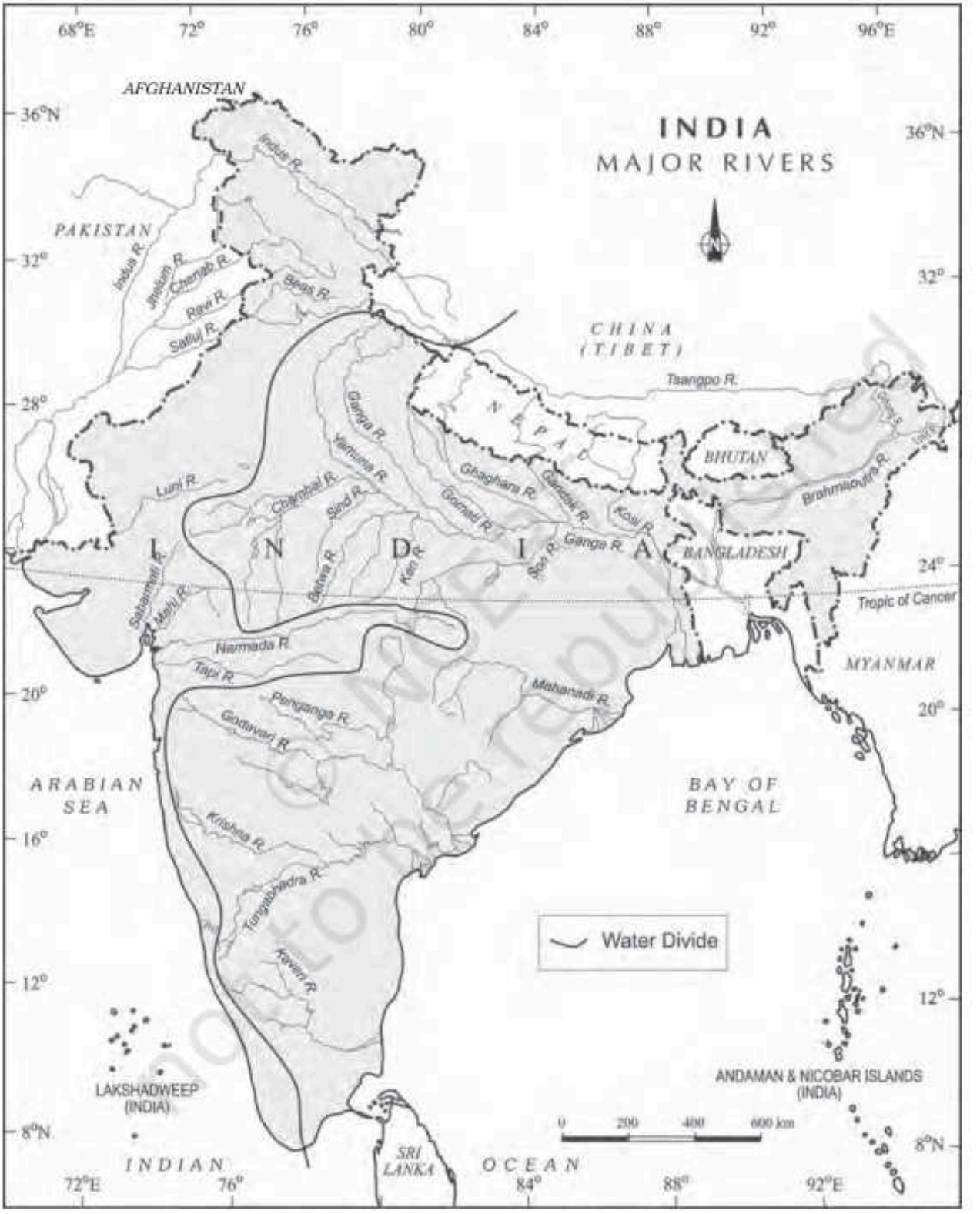


Figure 3.2 : Major Rivers of India

is called a drainage basin. The boundary line separating one drainage basin from the other is known as the watershed. The catchments of large rivers are called river basins while those of small rivulets and rills are often referred to as watersheds. There is, however, a slight difference between a river basin and a watershed. Watersheds are small in area while the basins cover larger areas.

River basins and watersheds are marked by unity. What happens in one part of the basin or watershed directly affects the other parts and the unit as a whole. That is why, they are accepted as the most appropriate micro, meso or macro planning regions.

Indian drainage system may be divided on various bases. On the basis of discharge of water (orientations to the sea), it may be grouped into: (i) the Arabian Sea drainage; and (ii) the Bay of Bengal drainage. They are separated from each other through the Delhi ridge, the Aravalis and the Sahyadris (water divide is shown by a line in Figure 3.1). Nearly 77 per cent of the drainage area consisting of the Ganga, the Brahmaputra, the Mahanadi, the Krishna, etc. is oriented towards the Bay of Bengal while 23 per cent comprising the Indus, the Narmada, the Tapi, the Mahi and the Periyar systems discharge their waters in the Arabian Sea.

On the basis of the size of the watershed, the drainage basins of India are grouped into three categories: (i) Major river basins with more than 20,000 sq. km of catchment area. It includes 14 drainage basins such as the Ganga, the Brahmaputra, the Krishna, the Tapi, the Narmada, the Mahi, the Pennar, the Sabarmati, the Barak, etc. (ii) Medium river basins with catchment area between 2,000-20,000 sq. km incorporating 44 river basins such as the Kalindi, the Periyar, the Meghna, etc. (iii) Minor river basins with catchment area of less than 2,000 sq. km include fairly good number of rivers flowing in the area of low rainfall.

If you look at the Figure 3.2 you can see that many rivers have their sources in the Himalayas and discharge their waters either in the Bay of Bengal or in the Arabian Sea. Identify these rivers of North India. Large rivers flowing on the Peninsular plateau have their origin in the Western Ghats and discharge their waters in the Bay of Bengal. Identify these rivers of the South India.

The Narmada and Tapi are two large rivers which are exceptions. They along with many small rivers discharge their waters in the Arabian Sea.

Name these rivers of the western coastal region from the Konkan to the Malabar coast.

On the basis of the mode of origin, nature and characteristics, the Indian drainage may also be classified into the Himalayan drainage and the Peninsular drainage. Although it has the problem of including the Chambal, the Betwa, the Son, etc. which are much older in age and origin than other rivers that have their origin in the Himalayas, it is the most accepted basis of classification. Hence, this scheme has been followed in this book.

Drainage Systems of India

Indian drainage system consists of a large number of small and big rivers. It is the outcome of the evolutionary process of the three major physiographic units and the nature and characteristics of precipitation.

The Himalayan Drainage

The Himalayan drainage system has evolved through a long geological history. It mainly includes the Ganga, the Indus and the Brahmaputra river basins. Since these are fed both by melting of snow and precipitation, rivers of this system are perennial. These rivers pass through the giant gorges carved out by the erosional activity carried on simultaneously with the uplift of the Himalayas. Besides deep gorges, these rivers also form V-shaped valleys,



Figure 3.3 : Rapids

course. While entering the plains, they form depositional features like flat valleys, ox-bow lakes, flood plains, braided channels, and deltas near the river mouth. In the Himalayan reaches, the course of these rivers is highly tortous, but over the plains they display a strong meandering tendency and shift their courses frequently. River Kosi, also know as the 'sorrow of Bihar', has been notorious for frequently changing its course. The Kosi brings huge quantity of sediments from its upper reaches and deposits it in the plains. The course gets blocked, and consequently, the river changes its course. Why does the Kosi river bring such huge quantity of sediments from the upper reaches? Do you think that the discharge of the water in the rivers in general and the Kosi in particular, remains the same, or does it fluctuate? When does the river course receive the maximum quantity of water? What are the positive and negative effects of flooding?

Evolution of the Himalayan Drainage

There are difference of opinion about the evolution of the Himalayan rivers. However, geologists believe that a mighty river called Shiwalik or Indo-Brahma traversed the entire longitudinal extent of the Himalaya from Assam to Punjab and onwards to Sind, and finally discharged into the Gulf of Sind near lower Punjab during the Miocene period some 5-24 million years ago The remarkable continuity of the Shiwalik and its lacustrine origin and alluvial deposits consisting of sands, silt, clay, boulders and conglomerates support this viewpoint.

It is opined that in due course of time Indo–Brahma river was dismembered into three main drainage systems: (i) the Indus and its five tributaries in the western part; (ii) the Ganga and its Himalayan tributaries in the central part; and (iii) the stretch of the Brahmaputra in Assam and its Himalayan tributaries in the eastern part. The dismemberment was probably due to the Pleistocene upheaval in the western Himalayas, including the uplift of the Potwar Plateau (Delhi Ridge), which acted as the water divide between the Indus and Ganga drainage systems. Likewise, the down-thrusting of the Malda gap area between the Rajmahal hills and the Meghalaya plateau during the mid-pleistocene period, diverted the Ganga and the Brahmaputra systems to flow towards the Bay of Bengal.

The River Systems of the Himalayan Drainage

The Himalayan drainage consists of several river systems but the following are the major river systems:

The Indus System

It is one of the largest river basins of the world, covering an area of 11,65,000 sq. km (in India it is 321, 289 sq. km and a total length of 2,880 km (in India 1,114 km). The Indus also known as the Sindhu, is the westernmost of the Himalayan rivers in India. It originates from a glacier near Bokhar Chu (31°°15' N latitude and 81°°40' E longitude) in the Tibetan region at an altitude of 4,164 m in the Kailash Mountain range. In Tibet, it is known as *'Singi Khamban*; or Lion's mouth. After flowing in the northwest direction between the Ladakh and Zaskar ranges, it passes through Ladakh and Baltistan. It cuts across the Ladakh range, forming a spectacular gorge near Gilgit in Ladakh. It enters into Pakistan near Chilas in the Dardistan region. Find out the area known as Dardistan.

The Indus receives a number of Himalayan tributaries such as the Shyok, the Gilgit, the Zaskar, the Hunza, the Nubra, the Shigar, the Gasting and the Dras. It finally emerges out of the hills near Attock where it receives the Kabul river on its right bank. The other important tributaries joining the right bank of the Indus are the Khurram, the Tochi, the Gomal, the Viboa and the Sangar. They all originate in the Sulaiman ranges. The river flows southward and receives 'Panjnad' a little above Mithankot. The Panjnad is the name given to the five rivers of Punjab, namely the Satluj, the Beas, the Ravi, the Chenab and the Jhelum. It finally

discharges into the Arabian Sea, east of Karachi. The Indus flows in India through Union Territories of Ladakh and Jammu and Kashmir.

The Jhelum, an important tributary of the Indus, rises from a spring at Verinag situated at the foot of the Pir Panjal in the south-eastern part of the valley of Kashmir. It flows through Srinagar and the Wular lake before entering Pakistan through a deep narrow gorge. It joins the Chenab near Jhang in Pakistan.

The Chenab is the largest tributary of the Indus. It is formed by two streams, the Chandra and the Bhaga, which join at Tandi near Keylong in Himachal Pradesh. Hence, it is also known as Chandrabhaga. The river flows for 1,180 km before entering into Pakistan.

The Ravi is another important tributary of the Indus. It rises west of the Rohtang pass in the Kullu hills of Himachal Pradesh and flows through the Chamba valley of the state. Before entering Pakistan and joining the Chenab near Sarai Sidhu, it drains the area lying between the southeastern part of the Pir Panjal and the Dhauladhar ranges.

The Beas is another important tributary of the Indus, originating from the Beas Kund near the Rohtang Pass at an elevation of 4,000 m above the mean sea level. The river flows through the Kullu valley and forms gorges at Kati and Largi in the Dhaoladhar range. It enters the Punjab plains where it meets the Satluj near Harike.

The Satluj originates in the 'Raksas tal' near Mansarovar at an altitude of 4,555 m in Tibet where it is known as Langchen Khambab. It flows almost parallel to the Indus for about 400 km before entering India, and comes out of a gorge at Rupar. It passes through the Shipki La on the Himalayan ranges and enters the Punjab plains. It is an antecedent river. It is a very important tributary as it feeds the canal system of the Bhakra Nangal project.

The Ganga System

The Ganga is the most important river of India both from the point of view of its basin and cultural significance. It rises in the Gangotri glacier near Gaumukh (3,900 m) in the Uttarkashi district of Uttarakhand. Here, it is known as the Bhagirathi. It cuts through the Central and the Lesser Himalayas in narrow gorges. At Devprayag, the Bhagirathi meets the Alaknanda; hereafter, it is known as the Ganga. The Alaknanda has its source in the Satopanth glacier above Badrinath. The Alaknanda consists of the Dhauli and the Vishnu Ganga which meet at Joshimath or Vishnu Prayag. The other tributaries of Alaknanda such as the Pindar joins it at Karna Prayag while Mandakini or Kali Ganga meets it at Rudra Prayag. The Ganga enters the plains at Haridwar. From here, it flows first to the south, then to the south-east and east before splitting into two distributaries, namely the Bhagirathi and the Padma. The river has a length of 2,525 km. It is shared by Uttarakhand (110 km) and Uttar Pradesh (1,450 km), Bihar (445 km) and West Bengal (520 km). The Ganga basin covers about 8.6 lakh sq. km area in India alone. The Ganga

Do you Know?

'Namami Gange Programme', is an Integrated Conservation Mission, approved as "Flagship Programme" by the Union Government in June 2014 with the twin objectives of effective abatement of pollution, conservation and rejuvenation of the National River Ganga.

Main pillars of the Namami Gange Programme are:

- Sewerage Treatment Infrastructure
- River-Front Development
- River-Surface Cleaning
- Bio-Diversity
- Afforestation
- Public Awareness
- Industrial Effluent Monitoring
- Ganga Gram

You may explore about this project at http://nmcg.nic.in/NamamiGanga.aspx#

river system is the largest in India having a number of perennial and non-perennial rivers originating in the Himalayas in the north and the Peninsula in the south, respectively. The Son is its major right bank tributary. The important left bank tributaries are the Ramganga, the Gomati, the Ghaghara, the Gandak, the Kosi and the Mahananda. The river finally discharges itself into the Bay of Bengal near the Sagar Island.

The Yamuna, the western most and the longest tributary of the Ganga, has its source in the Yamunotri glacier on the western slopes of Banderpunch range (6,316 m). It joins the Ganga at Prayag (Allahabad). It is joined by the Chambal, the Sind, the Betwa and the Ken on its right bank which originates from the Peninsular plateau while the Hindan, the Rind, the Sengar, the Varuna, etc. join it on its left bank. Much of its water feeds the western and eastern Yamuna and the Agra canals for irrigation purposes.

Name the states which are drained by the river Yamuna.

The Chambal rises near Mhow in the Malwa plateau of Madhya Pradesh and flows northwards through a gorge up wards of Kota in Rajasthan, where the Gandhisagar dam has been constructed. From Kota, it traverses down to Bundi, Sawai Madhopur and Dholpur, and finally joins the Yamuna. The Chambal is famous for its badland topography called the Chambal ravines.

The Gandak comprises two streams, namely Kaligandak and Trishulganga. It rises in the Nepal Himalayas between the Dhaulagiri and Mount Everest and drains the central part of Nepal. It enters the Ganga plain in Champaran district of Bihar and joins the Ganga at Sonpur near Patna.

The Ghaghara originates in the glaciers of Mapchachungo. After collecting the waters of its tributaries – Tila, Seti and Beri, it comes out of the mountain, cutting a deep gorge at Shishapani. The river Sarda (Kali or Kali Ganga) joins it in the plain before it finally meets the Ganga at Chhapra.

The Kosi is an antecedent river with its source to the north of Mount Everest in Tibet, where its main stream Arun rises. After crossing the Central Himalayas in Nepal, it is joined by the Son Kosi from the West and the Tamur Kosi from the east. It forms Sapt Kosi after uniting with the river Arun.

The Ramganga is comparatively a small river rising in the Garhwal hills near Gairsain. It changes its course to the southwest direction after crossing the Shiwalik and enters into the plains of Uttar Pradesh near Najibabad. Finally, it joins the Ganga near Kannauj.

The Damodar occupies the eastern margins of the Chotanagpur Plateau where it flows through a rift valley and finally joins the Hugli. The Barakar is its main tributary. Once known as the 'sorrow of Bengal', the Damodar has been now tamed by the Damodar Valley corporation, a multipurpose project.

The Sarda or Saryu river rises in the Milam glacier in the Nepal Himalayas where it is known as the Goriganga. Along the Indo-Nepal border, it is called Kali or Chauk, where it joins the Ghaghara.

The Mahananda is another important tributary of the Ganga rising in the Darjiling hills. It joins the Ganga as its last left bank tributary in West Bengal.

The Son is a large south bank tributary of the Ganga, originating in the Amarkantak plateau. After forming a series of waterfalls at the edge of the plateau, it reaches Arrah, west of Patna, to join the Ganga.

The Brahmaputra System

The Brahmaputra, one of the largest rivers of the world, has its origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake. From here, it traverses eastward longitudinally for a distance of nearly 1,200 km in a dry and flat region of southern Tibet, where it is known as the Tsangpo, which means 'the purifier.' The Rango Tsangpo is the major right bank tributary of this river in Tibet. It emerges as a turbulent and dynamic river after carving out a deep gorge in the Central

Himalayas near Namcha Barwa (7,755 m). The river emerges from the foothills under the name of Siang or Dihang. It enters India west of Sadiya town in Arunachal Pradesh. Flowing southwest, it receives its main left bank tributaries, viz., Dibang or Sikang and Lohit; thereafter, it is known as the Brahmaputra.

The Brahmaputra receives numerous tributaries in its 750 km long journey through the Assam valley. Its major left bank tributaries are the Burhi Dihing and Dhansari (South) whereas the important right bank tributaries are the Subansiri, Kameng, Manas and Sankosh. The Subansiri which has its origin in Tibet, is an antecedent river. The Brahmaputra enters into Bangladesh near Dhubri and flows southward. In Bangladesh, the Tista joins it on its right bank from where the river is known as the Jamuna. It finally merges with the river Padma, which falls in the Bay of Bengal. The Brahmaputra is well-known for floods, channel shifting and bank erosion. This is due to the fact that most of its tributaries are large, and bring large quantity of sediments owing to heavy rainfall in its catchment area.

The Peninsular Drainage System

The Peninsular drainage system is older than the Himalayan one. This is evident from the broad, largely-graded shallow valleys, and the maturity of the rivers. The Western Ghats running close to the western coast act as the water divide between the major Peninsular rivers, discharging their water in the Bay of Bengal and as small rivulets joining the Arabian Sea. Most of the major Peninsular rivers except Narmada and Tapi flow from west to east. The Chambal, the Sind, the Betwa, the Ken, the Son, originating in the northern part of the Peninsula belong to the Ganga river system. The other major river systems of the Peninsular drainage are – the Mahanadi the Godavari, the Krishna and the Kaveri. Peninsular rivers are characterised by fixed course, absence of meanders and non-perennial flow of water. The Narmada and the Tapi which flow through the rift valley are, however, exceptions.

The Evolution of Peninsular Drainage System

Three major geological events in the distant past have shaped the present drainage systems of Peninsular India: (i) Subsidence of the western flank of the Peninsula leading to its submergence below the sea during the early tertiary period. Generally, it has disturbed the symmetrical plan of the river on either side of the original watershed. (ii) Upheaval of the Himalayas when the northern flank of the Peninsular block was subjected to subsidence and the consequent trough faulting. The Narmada and The Tapi flow in trough faults and fill the original cracks with their detritus materials. Hence, there is a lack of alluvial and deltaic deposits in these rivers. (iii) Slight tilting of the Peninsular block from northwest to the southeastern direction gave orientation to the entire drainage system towards the Bay of Bengal during the same period.

River Systems of the Peninsular Drainage

There are a large number of river systems in the Peninsular drainage. A brief account of the major Peninsular river systems is given below:

The Mahanadi rises near Sihawa in Raipur district of Chhattisgarh and runs through Odisha to discharge its water into the Bay of Bengal. It is 851 km long and its catchment area spreads over 1.42 lakh sq. km. Some navigation is carried on in the lower course of this river. Fifty three per cent of the drainage basin of this river lies in Madhya Pradesh and Chhattisgarh, while 47 per cent lies in Odisha.

The Godavari is the largest Peninsular river system. It is also called the Dakshin Ganga. It rises in the Nasik district of Maharashtra and discharges its water into the Bay of Bengal. Its tributaries run through the states of Maharashtra, Madhya Pradesh, Chhattisgarh, Odisha and Andhra Pradesh. It is 1,465 km long with a catchment area spreading over 3.13 lakh sq. km 49 per cent of this, lies in Maharashtra, 20 per cent in Madhya Pradesh and Chhattisgarh, and the rest in Andhra Pradesh. The Penganga, the Indravati, the

Pranhita, and the Manjra are its principal tributaries. The Godavari is subjected to heavy floods in its lower reaches to the south of Polavaram, where it forms a picturesque gorge. It is navigable only in the deltaic stretch. The river after Rajamundri splits into several branches forming a large delta.

The Krishna is the second largest east-flowing Peninsular river which rises near Mahabaleshwar in Sahyadri. Its total length is 1,401 km. The Koyna, the Tungbhadra and the Bhima are its major tributaries. Of the total catchment area of the Krishna, 27 per cent lies in Maharashtra, 44 per cent in Karnataka and 29 per cent in Andhra Pradesh and Telangana.

The Kaveri rises in Brahmagiri hills (1,341m) of Kogadu district in Karnataka. Its length is 800 km and it drains an area of 81,155 sq. km. Since the upper catchment area receives rainfall during the southwest monsoon season (summer) and the lower part during the northeast monsoon season (winter), the river carries water throughout the year with comparatively less fluctuation than the other Peninsular rivers. About 3 per cent of the Kaveri basin falls in Kerala, 41 per cent in Karnataka and 56 per cent in Tamil Nadu. Its important tributaries are the Kabini, the Bhavani and the Amravati.

The Narmada originates on the western flank of the Amarkantak plateau at a height of about 1,057 m. Flowing in a rift valley between the Satpura in the south and the Vindhyan range in the north, it forms a picturesque gorge in marble rocks and Dhuandhar waterfall near Jabalpur. After flowing a distance of about 1,312 km, it meets the Arabian sea south of Bharuch, forming a broad 27 km long estuary. Its catchment area is about 98,796 sq. km. The Sardar Sarovar Project has been constructed on this river.

Collect information about Narmada river conservation mission named "Namami Devi Narmade" and discuss with your peers.

The Tapi is the other important westward flowing river. It originates from Multai in the Betul district of Madhya Pradesh. It is 724 km long and drains an area of 65,145 sq. km. Nearly 79 per cent of its basin lies in Maharashtra, 15 per cent in Madhya Pradesh and the remaining 6 per cent in Gujarat.

Luni is the largest river system of Rajasthan, west of Aravali. It originates near Pushkar in two branches, i.e. the Saraswati and the Sabarmati, which join with each other at Govindgarh. From here, the river comes out of Aravali and is known as Luni. It flows towards the west till Telwara and then takes a southwest direction to join the Rann of Kuchchh. The entire river system is ephemeral.

Extent of Usability of River Water

The rivers of India carry huge volumes of water per year but it is unevenly distributed both in time and space. There are perennial rivers carrying water throughout the year while the non-perennial rivers have very little water during the dry season. During the rainy season, much of the water is wasted in floods and flows down to the sea. Similarly, when there is a flood in one part of the country, the other area suffers from drought. Why does this happen? Is it the problem of availability of water resource or that of its management? Can you suggest some measures to mitigate the problems of floods and droughts simultaneously occuring in different parts of the country? (See Chapter 6 of the book).

Can these problems be solved or minimised by trasfering the surplus water from one basin to the water deficit basins? Do we have some schemes of inter-basin linkage?

Teachers may explain the following examples

- Periyar Diversion Scheme
- Indira Gandhi Canal Project
- Kurnool-Cuddapah Canal
- Beas-Satluj Link Canal
- Ganga-Kaveri Link Canal